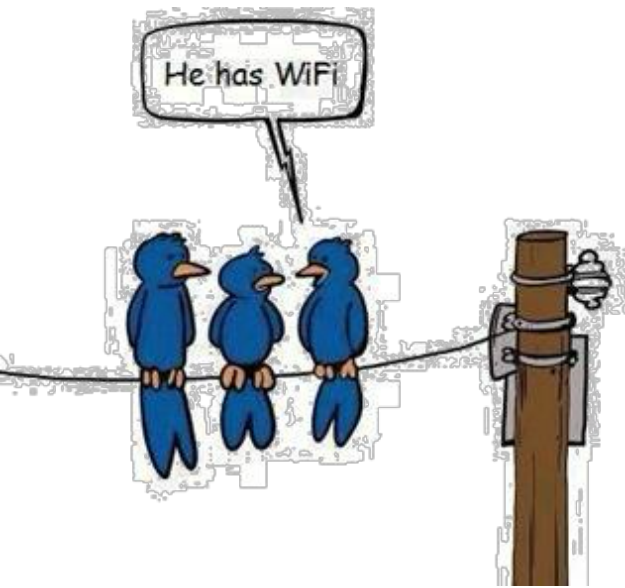


ECSE 4560/ 6560

Modern Communication Systems

Lecture 9: OFDM Orthogonality



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Orthogonality**
- Projects
 - Project Topics and Template posted on Piazza
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due Friday **10/3**
 - **1 page: Project Title, Abstract, and Evaluation Plan**
 - **Extra office hour?**
- HW2/Lab2 due Mon 9/29 (extended Wed 10/1)
 - Need extension?
- HW3/Lab3 will be posted today.

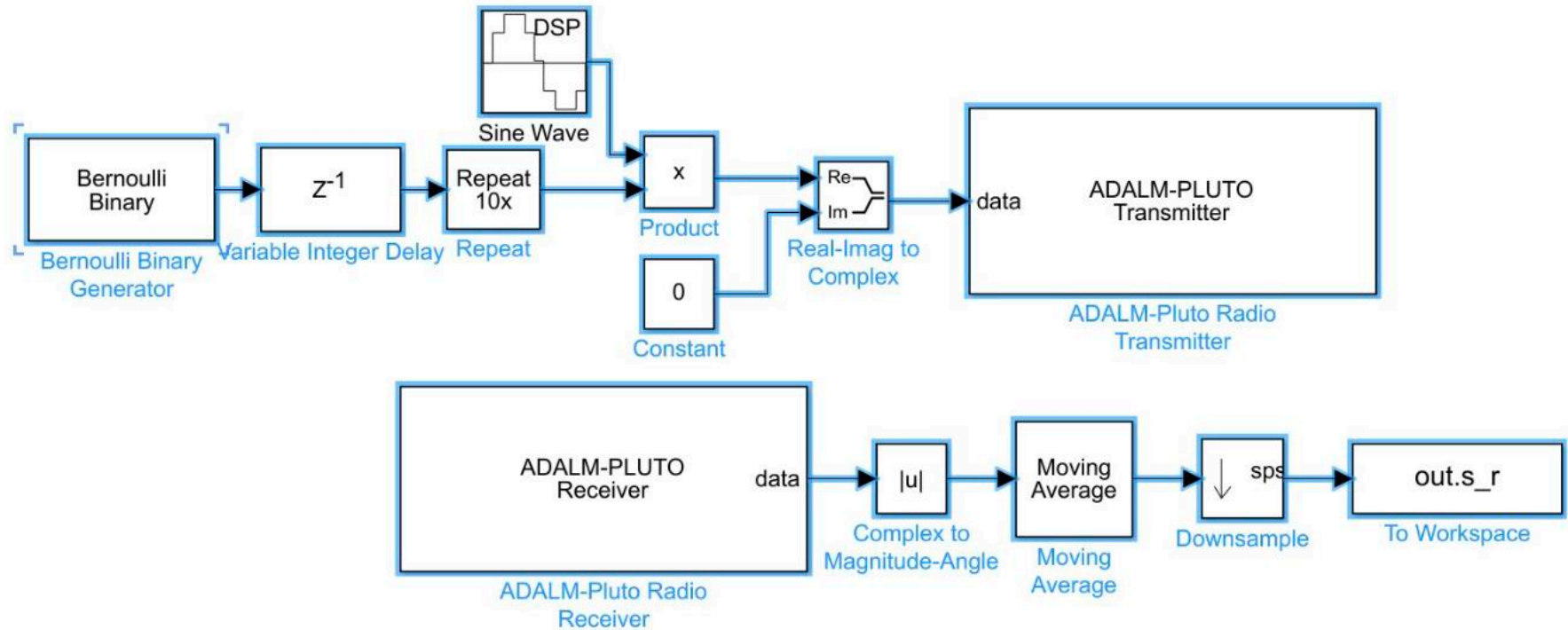
Today's Class: Multipath Channel (Cont.)+OFDM

- Multipath Channel (Revision)
- OFDM
 - Orthogonality
 - DFT Encoding
 - Cyclic Prefix
 - IDFT Decoding
 - Packet detection

Hardware Pluto+ SDR

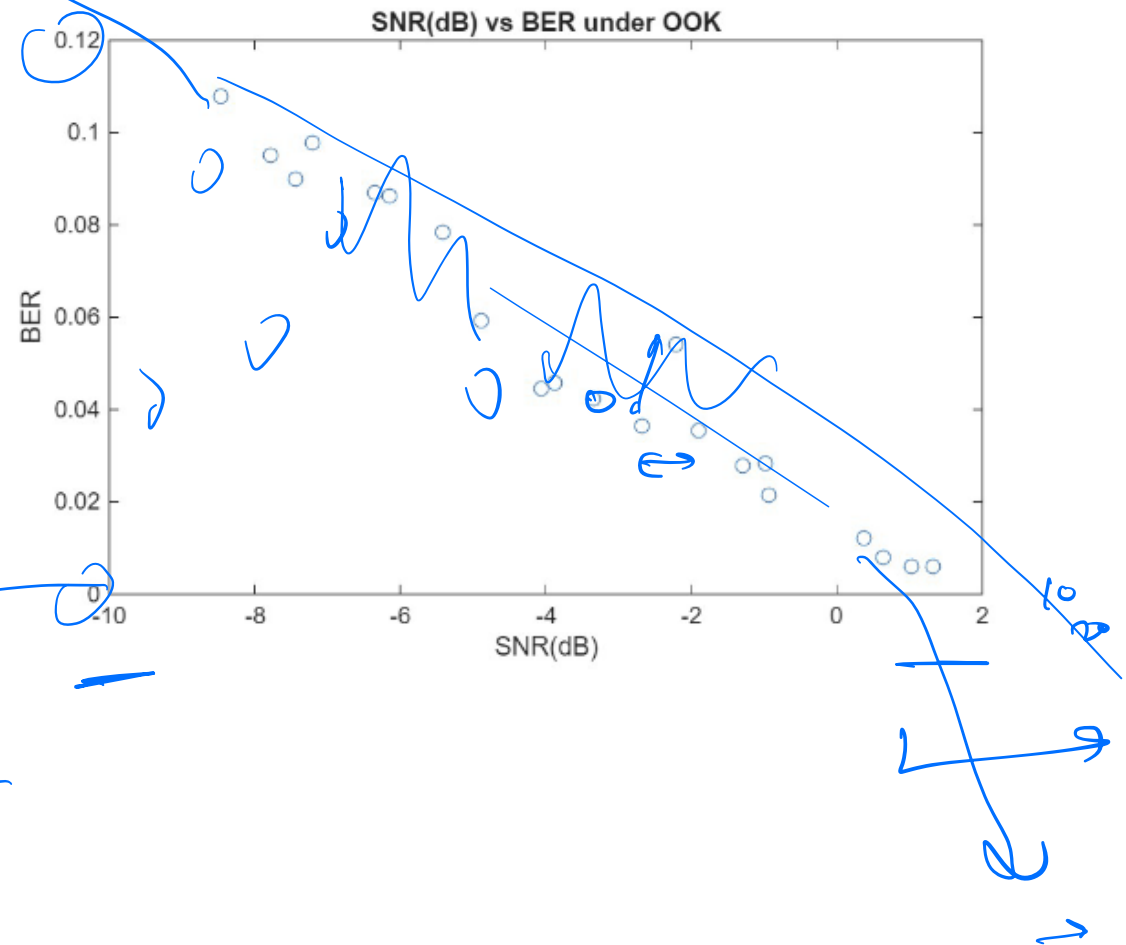
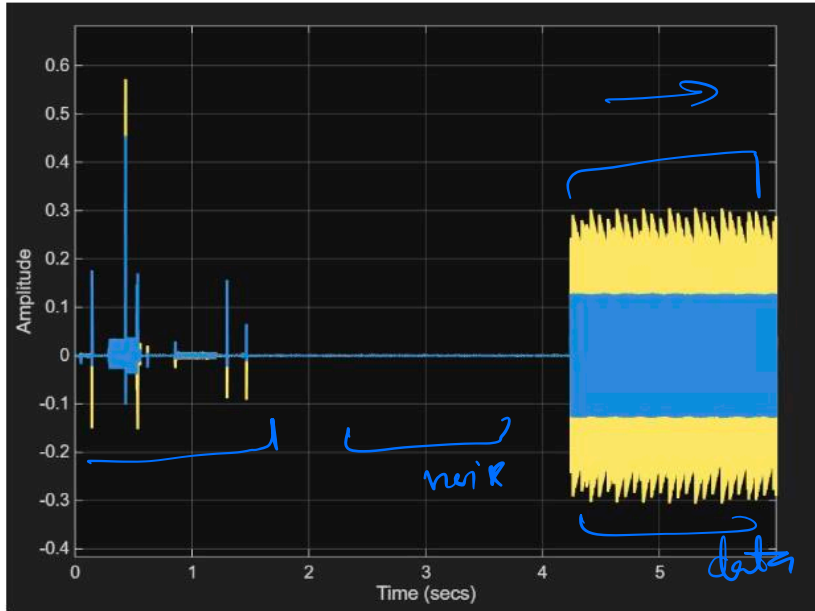


Hardware Pluto+ SDR

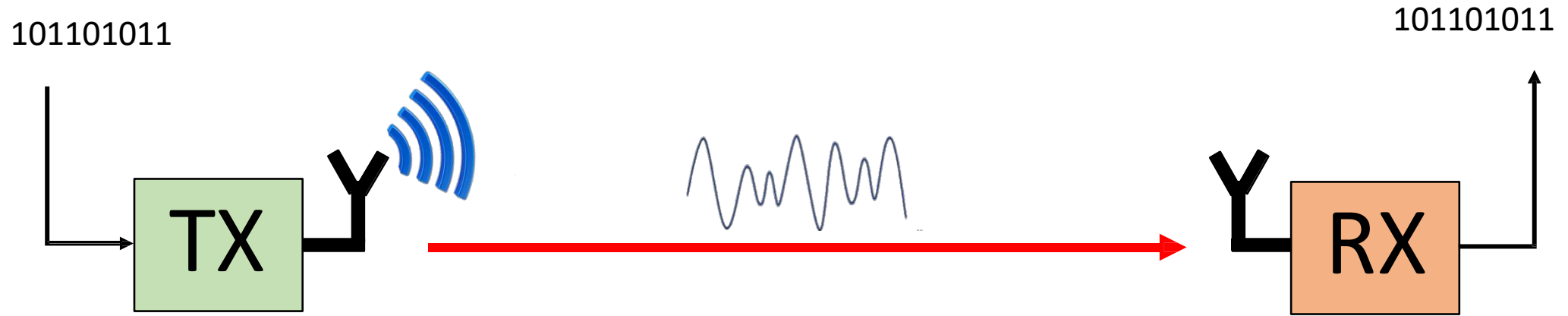


Hardware Pluto+ SDR

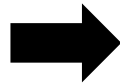
$$Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$



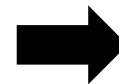
The Channel



$x(t)$



Channel

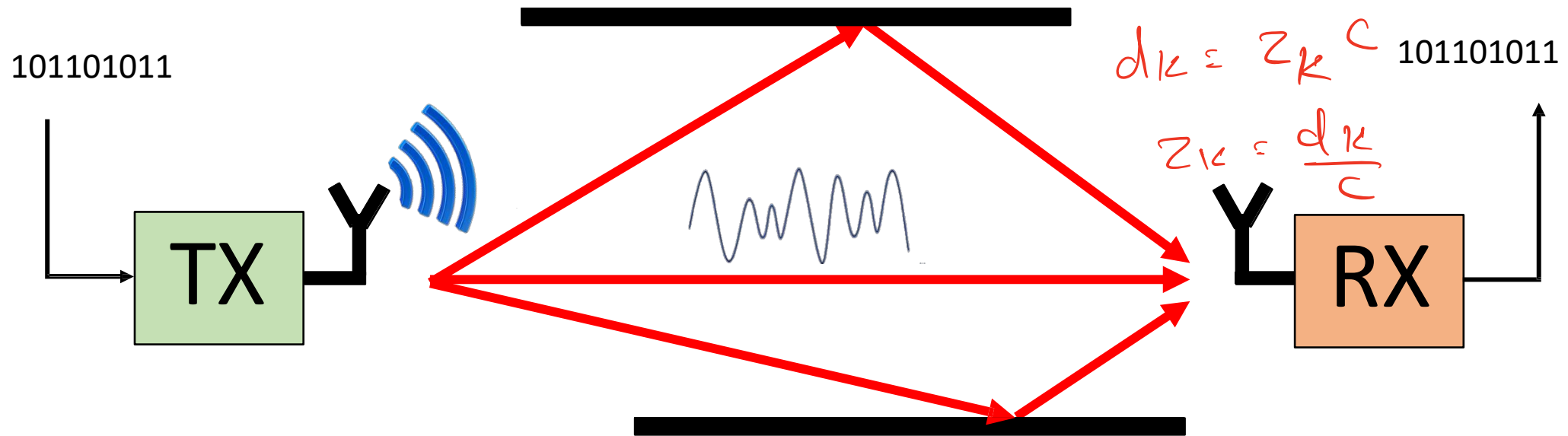


$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

Assumes single path!

Multipath Channel



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times
 ← phase

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

atten. delay → multiple copies

$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

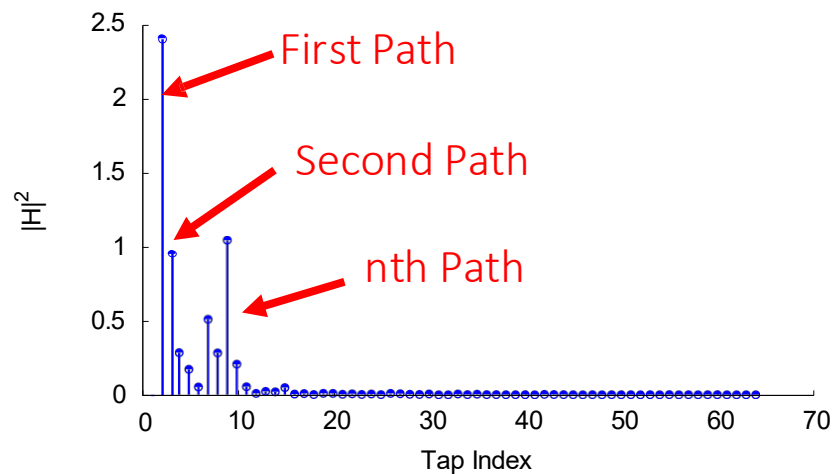
$$\phi \propto \frac{2\pi d_k}{\lambda}$$

$h(t)$ is channel impulse response.

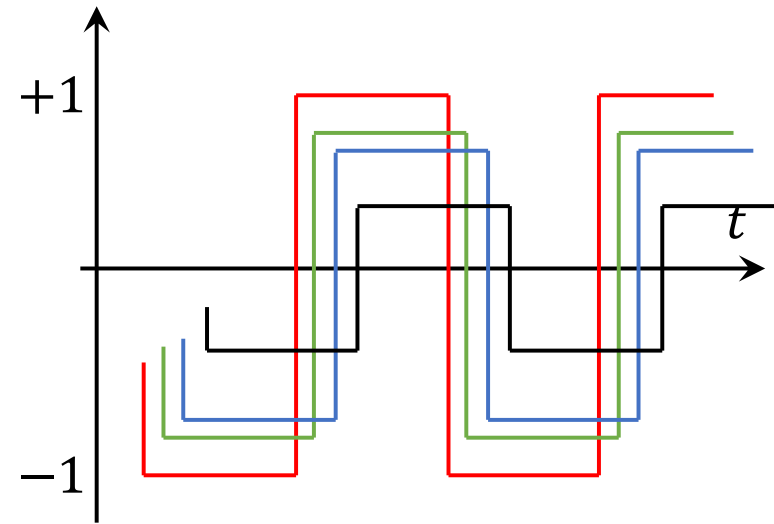
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



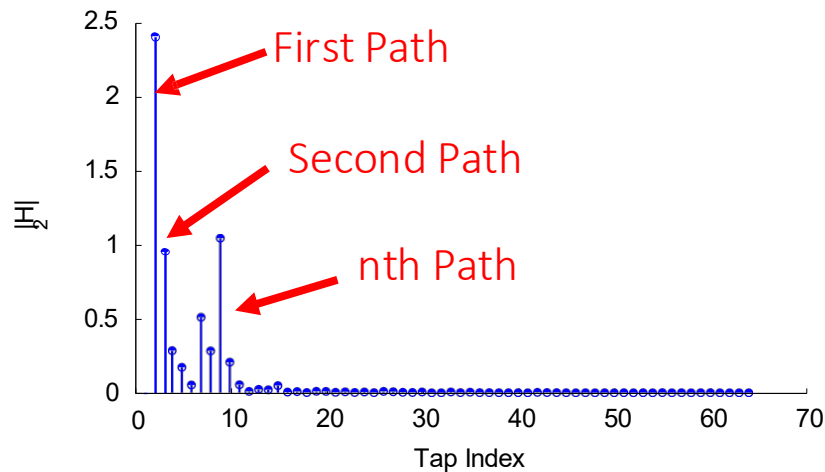
ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.

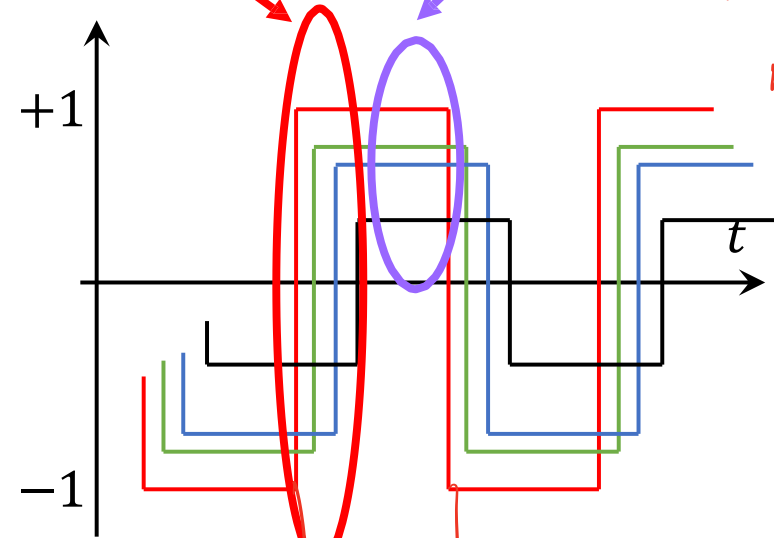
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.

Channel Fading

Symbols arriving along different paths sum up destructively

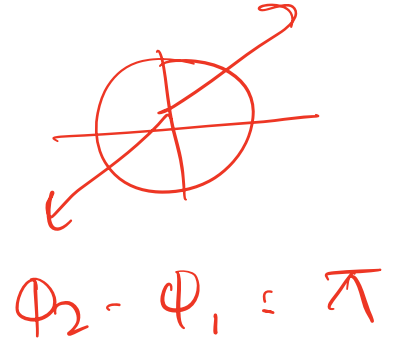
bad

Paths sum with different phases:
Constructive/Destructive

equalize not bad

Multipath Channel

$h(t)$ is channel impulse response.



$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi(d_2-d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1$$

$$\text{if } \frac{d_2 - d_1}{\lambda} \approx \frac{1}{2} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + e^{j\pi}) = 0$$

Destructive Interference!

Multipath Channel

$h(t)$ is channel impulse response.

$$\Phi_k = 2\pi \frac{d_k}{\lambda}$$

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

@ $f_1 = 2.5GHz$ ($\lambda = 12\text{ cm}$): $h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$

@ $f_2 = 5GHz$ ($\lambda = 6\text{ cm}$): $h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$



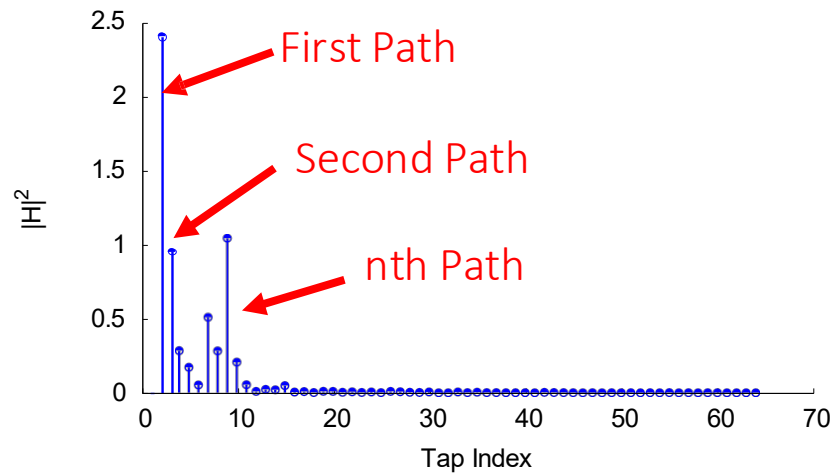
17x
(24dB)

Frequency Selective Fading

Multipath Channel

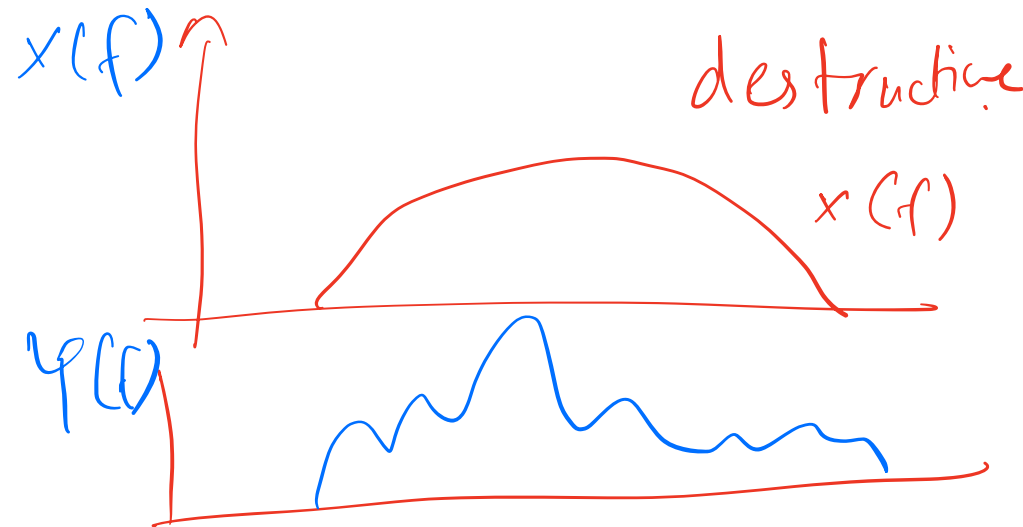
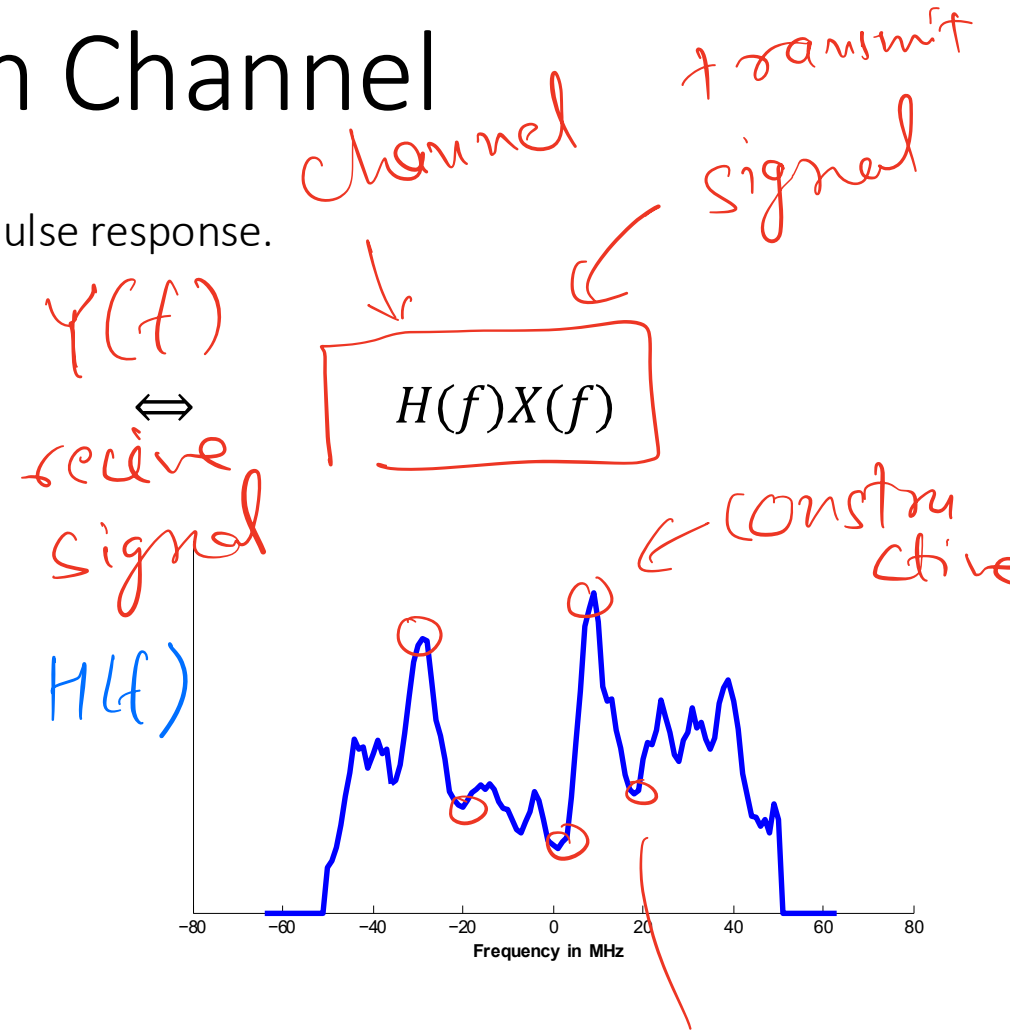
$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel

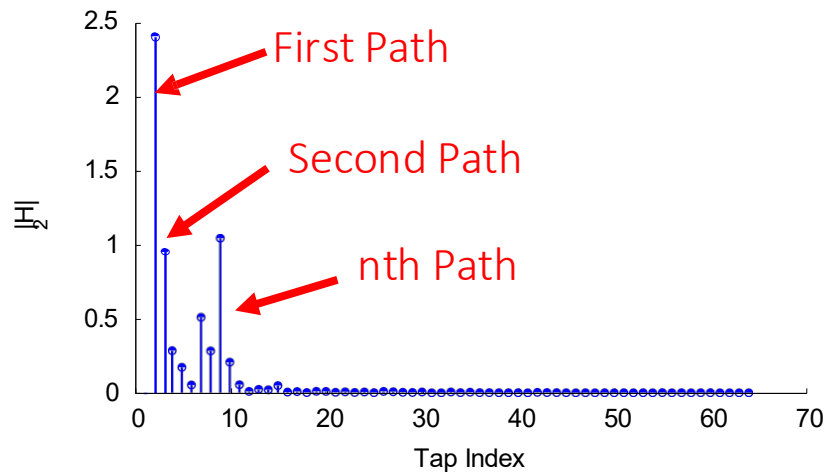
freq. selective fading



Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$

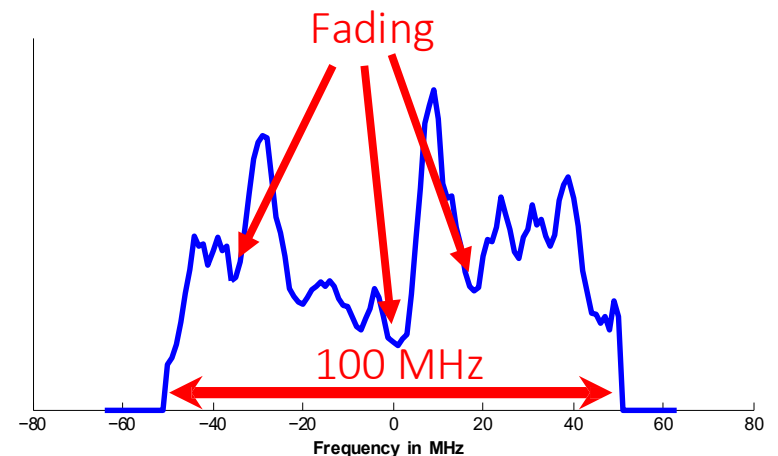


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

Symbols arriving along different paths sum up destructively

Problematic in
Wideband Channel!

Narrowband Channel

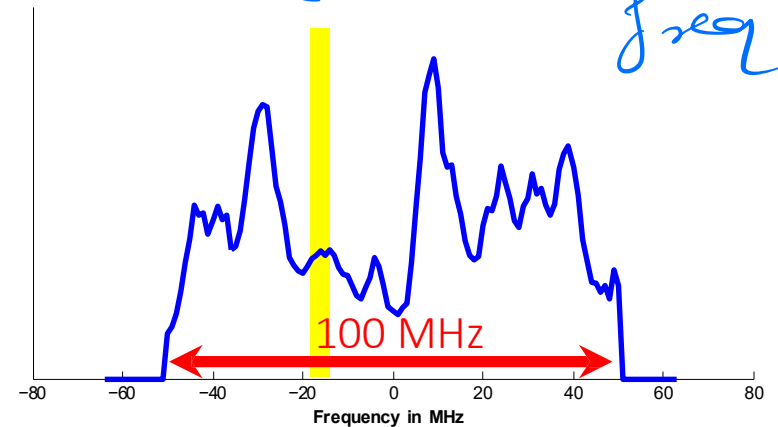
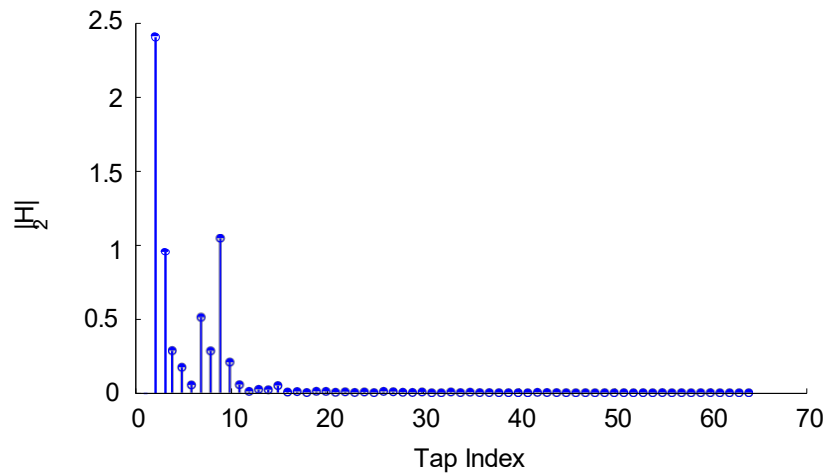
$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$\Leftrightarrow$

$$H(f)X(f)$$

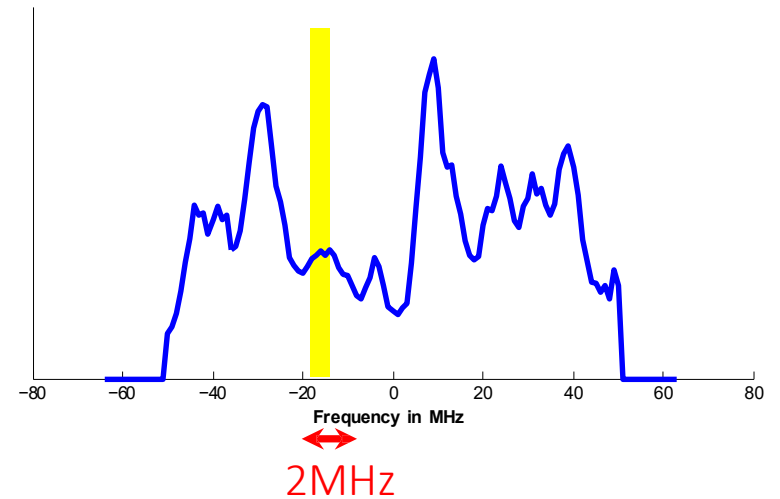
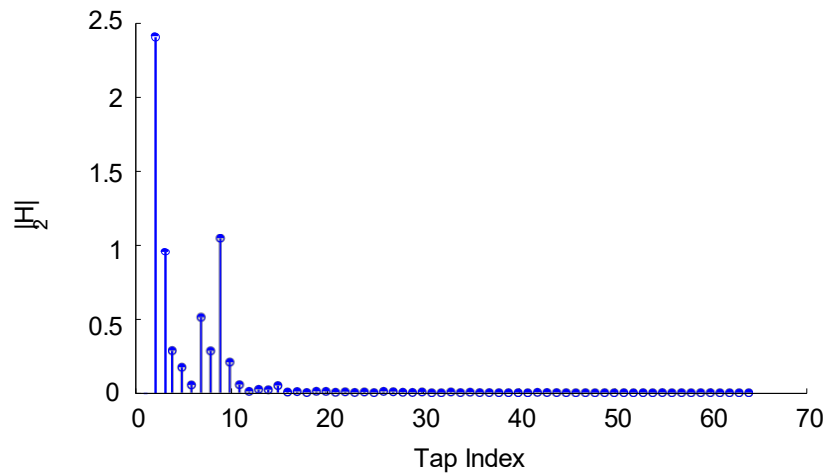
$\Leftrightarrow$ slice of the freq



Narrowband Channel

$h(t)$ is channel impulse response.

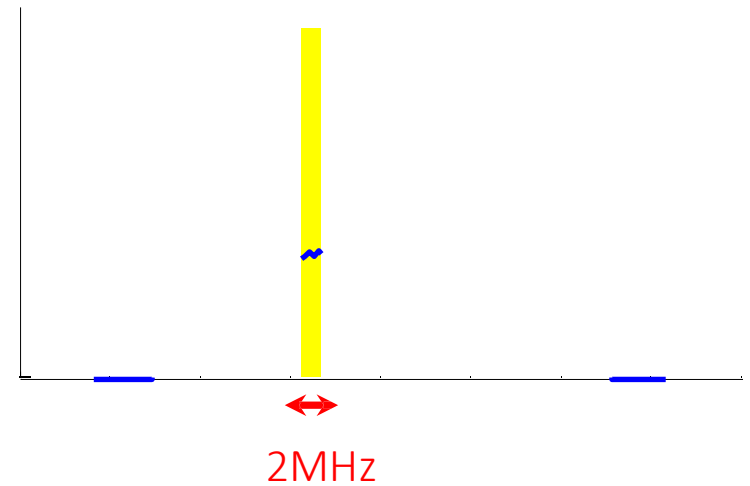
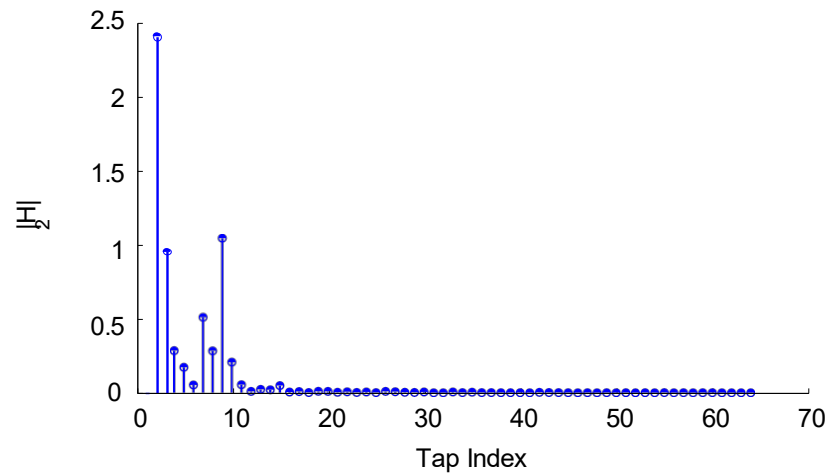
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

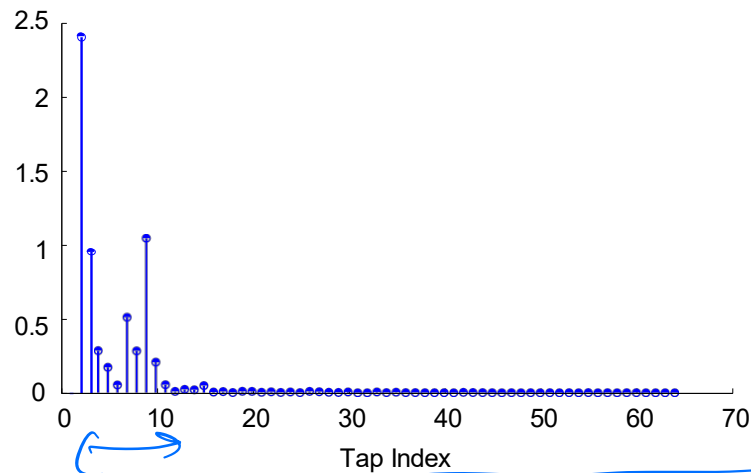
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

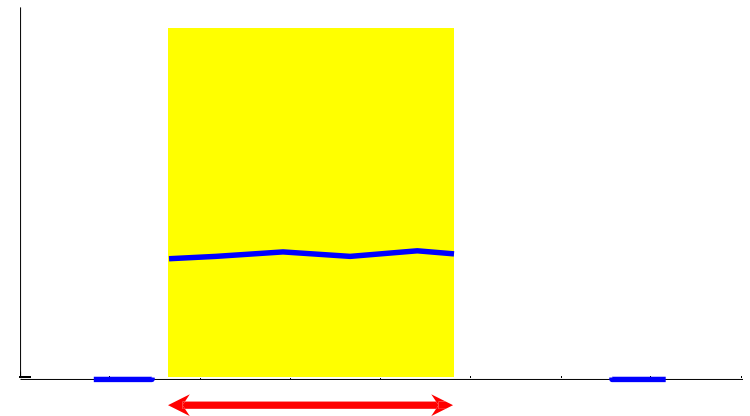
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



delay spread
2k

Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$

$$f_s = \frac{1}{T}$$



Narrowband

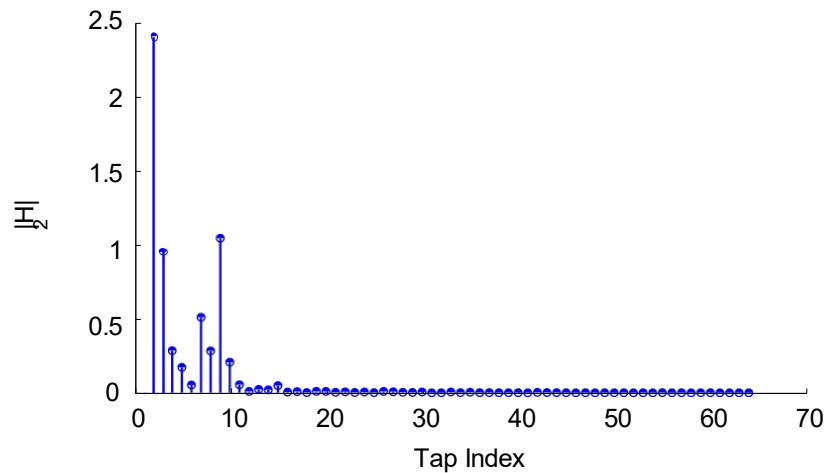
Flat Channel

2 MHz

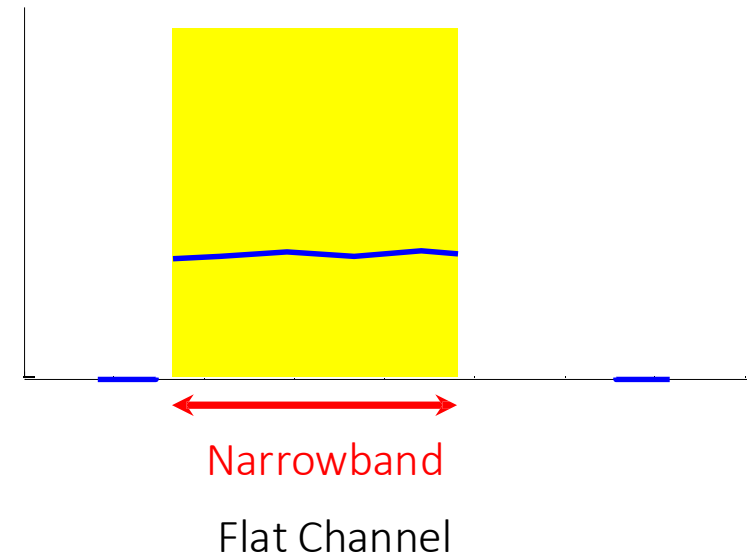
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$

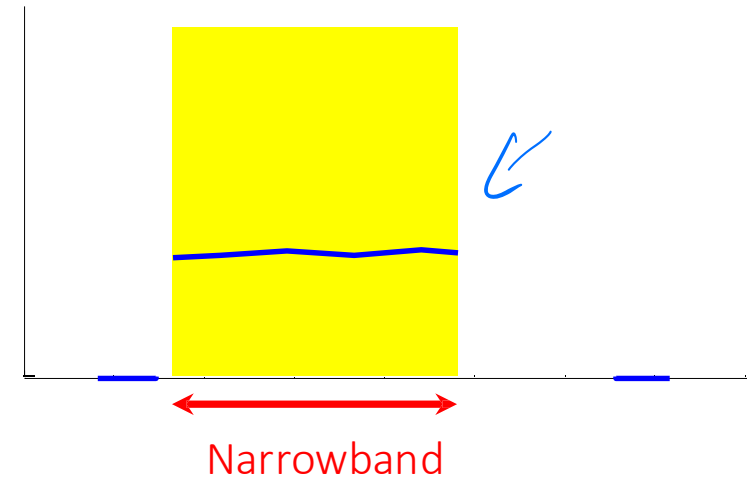
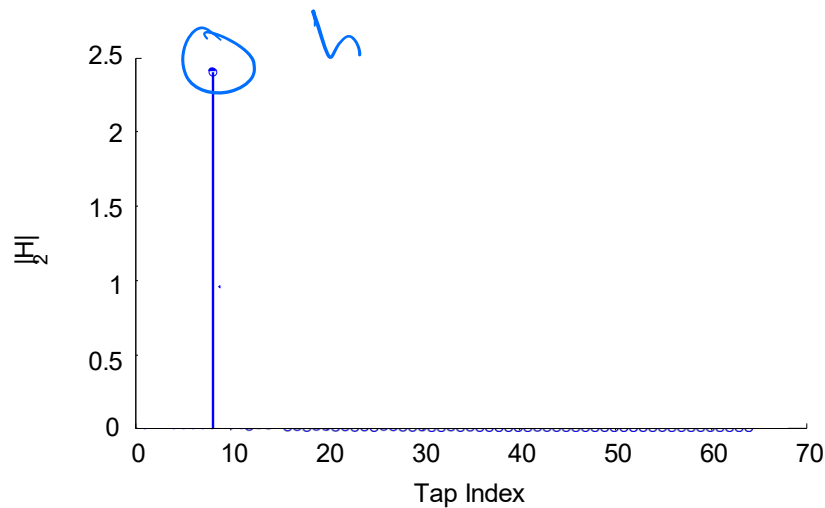


$\Rightarrow$ NO ISI
no freq. selective channel
but, low Bw, low data rates

Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$

$\gg \tau_k$

Flat Channel

$x(t - \tau_k) \approx x(t)$

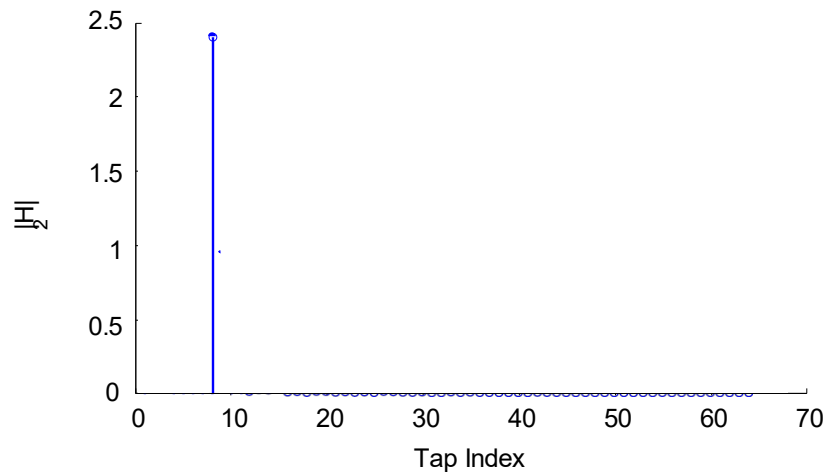
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

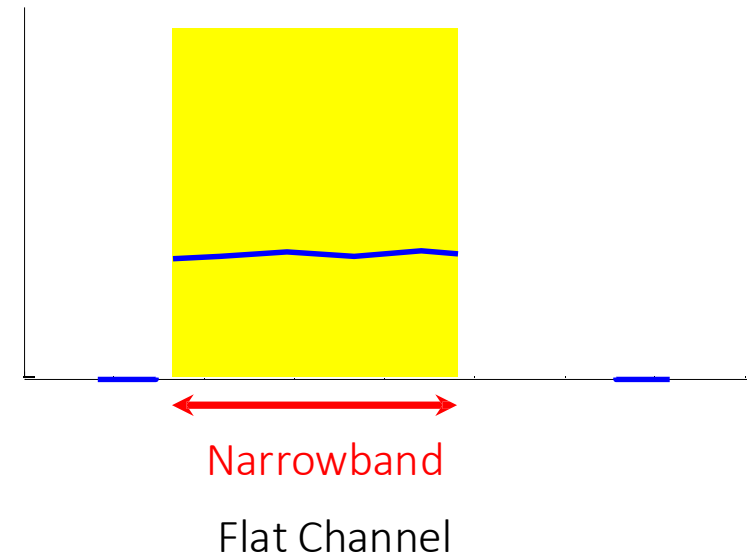
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



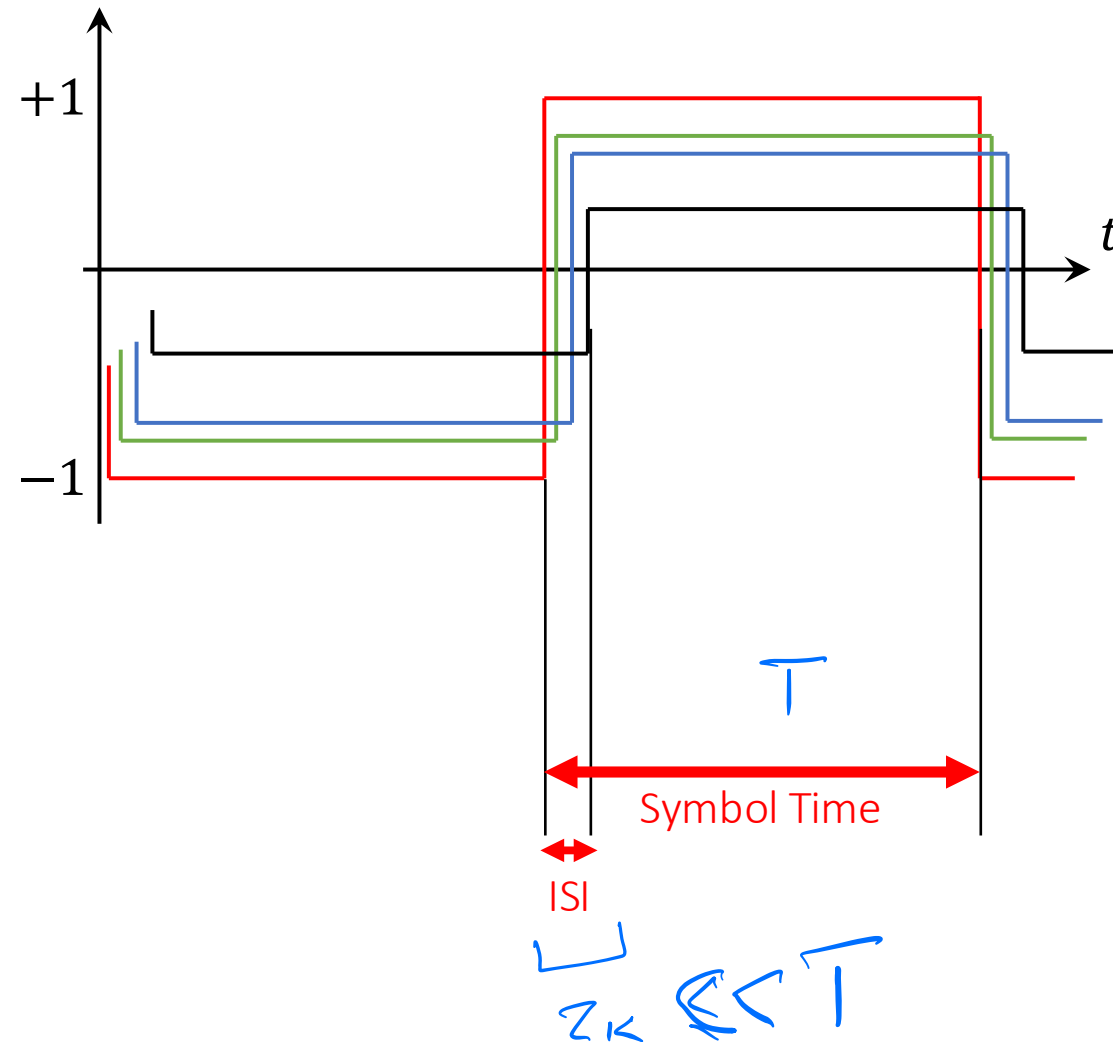
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = h x(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

Narrowband vs. Wideband Channel

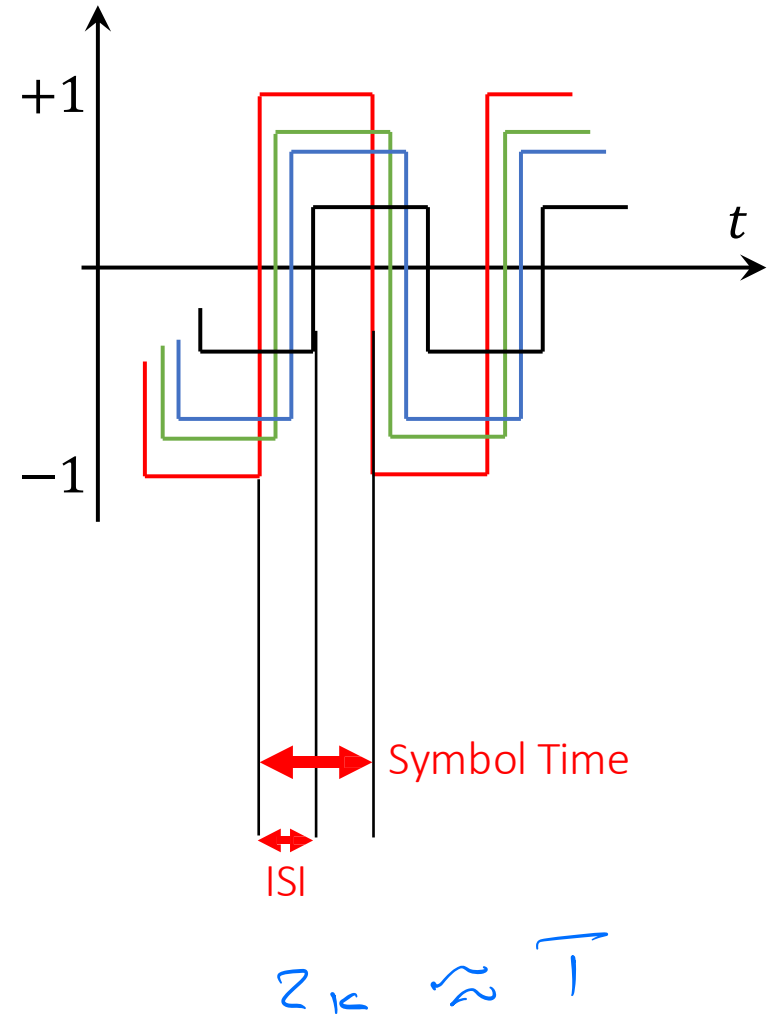
Narrowband Channel

$$h x(t)$$



Wideband Channel

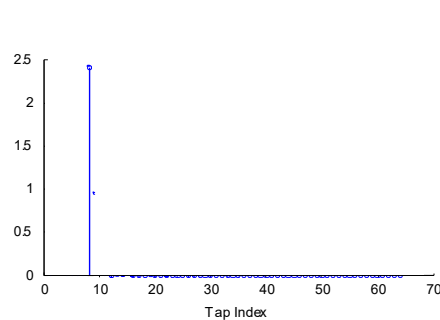
$$h(t) * x(t)$$



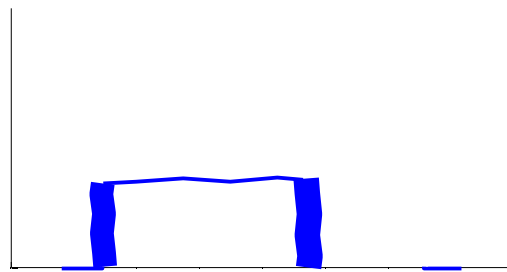
Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t) \iff h X(f)$$



$\approx$ Single Tap



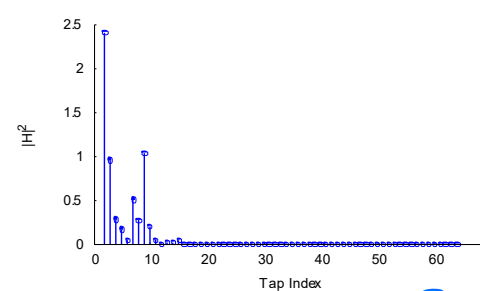
$\approx$ Flat Channel

$$C = B \log(1 + \text{SNR})$$

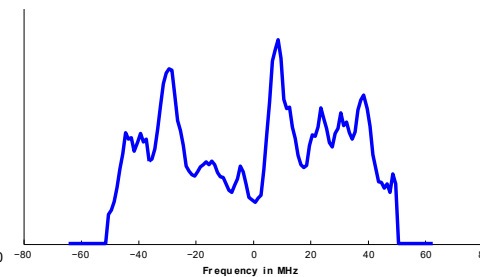
high BW
 $\Rightarrow$ high data rates

Wideband Channel

$$h(t) * x(t) \iff H(f) X(f)$$



Multi-tap
Channel
(ISI)

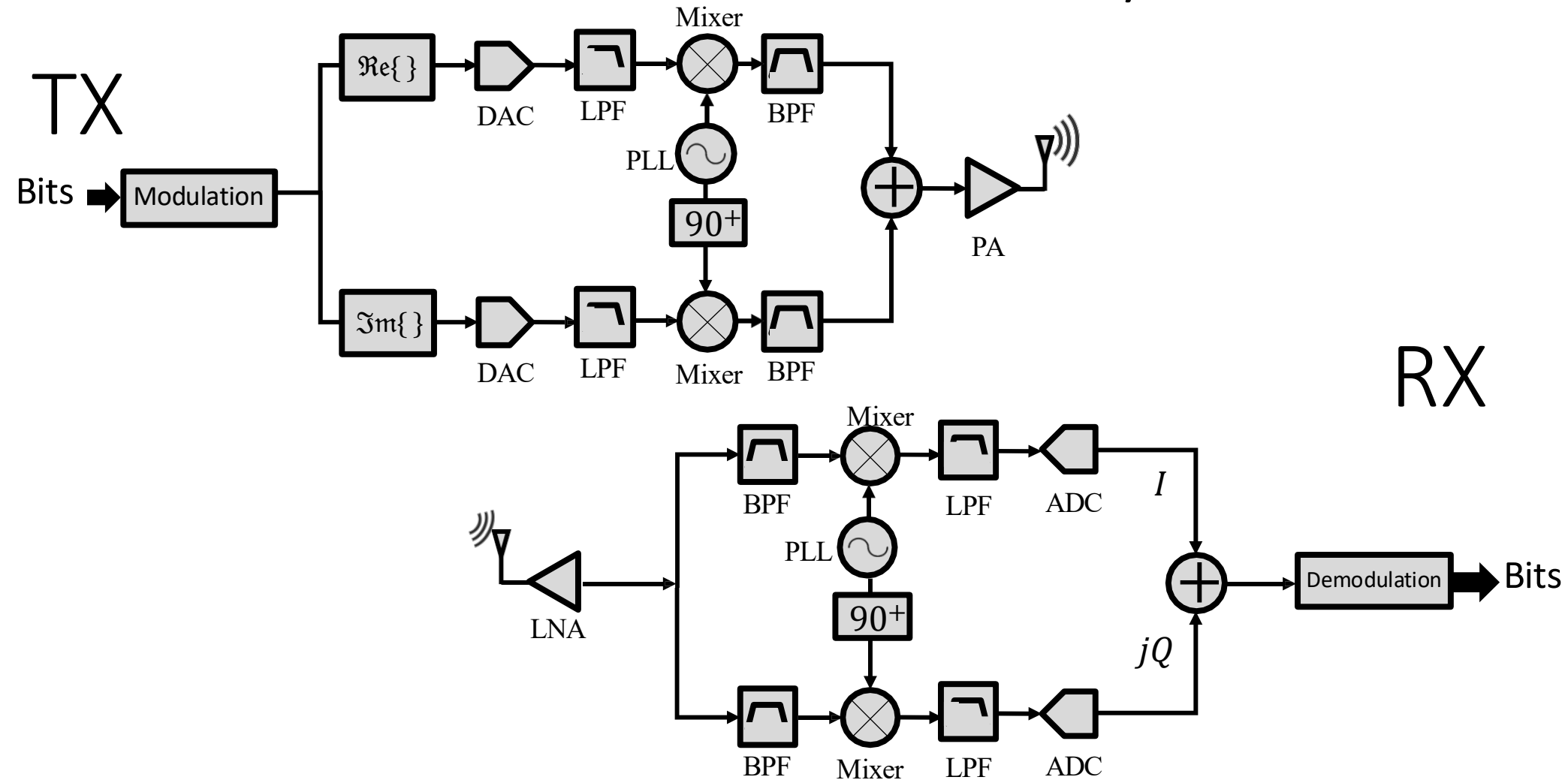


Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: OFDM

Wireless Communication System



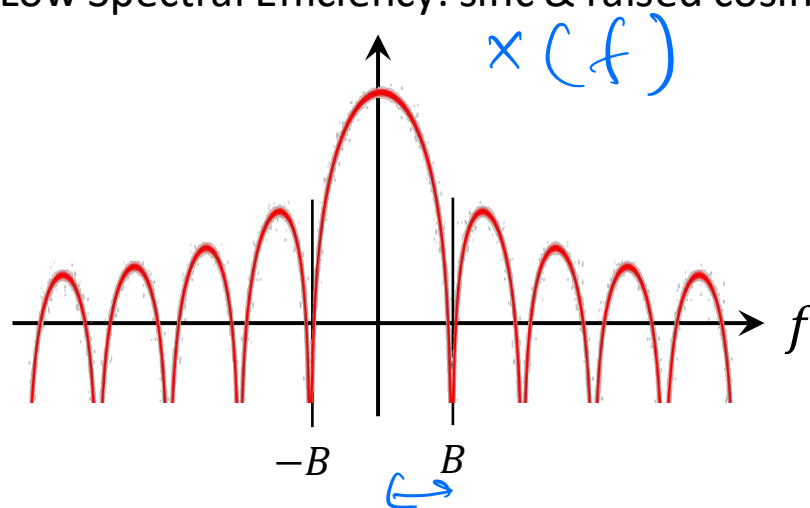
Single Carrier Modulation

Symbols modulated on a single carrier frequency: $\underline{s[n]} e^{-j2\pi f_c t}$

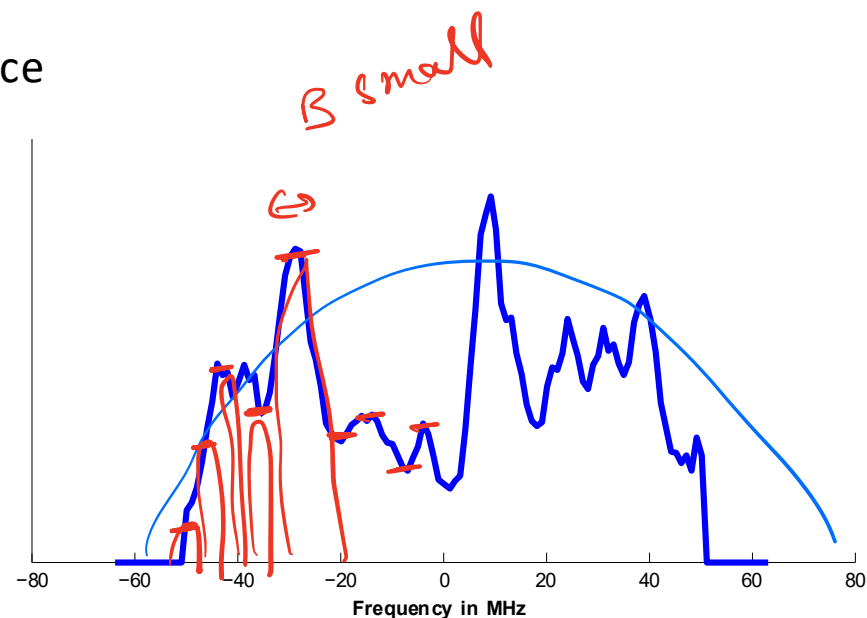
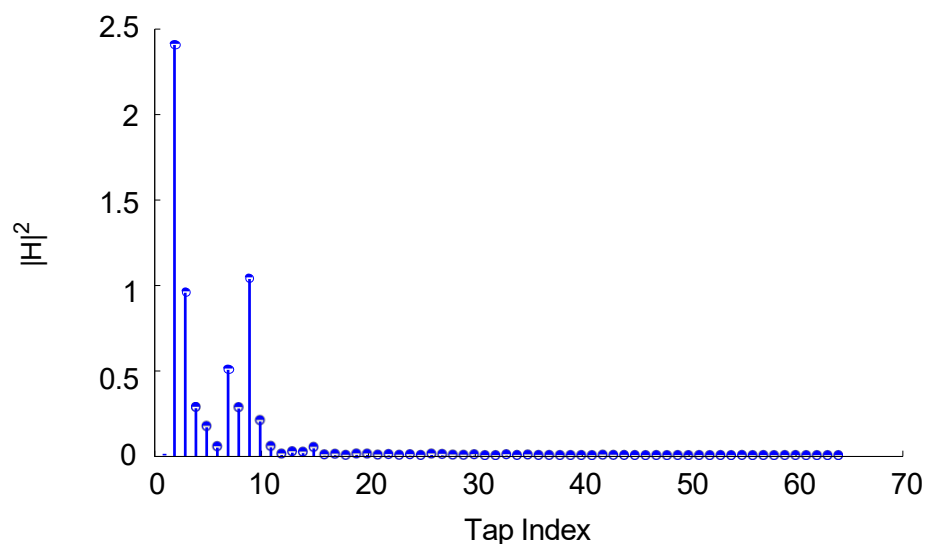
Single Carrier Modulation

Symbols modulated on a single carrier frequency

- Low Spectral Efficiency: sinc & raised cosine leakage



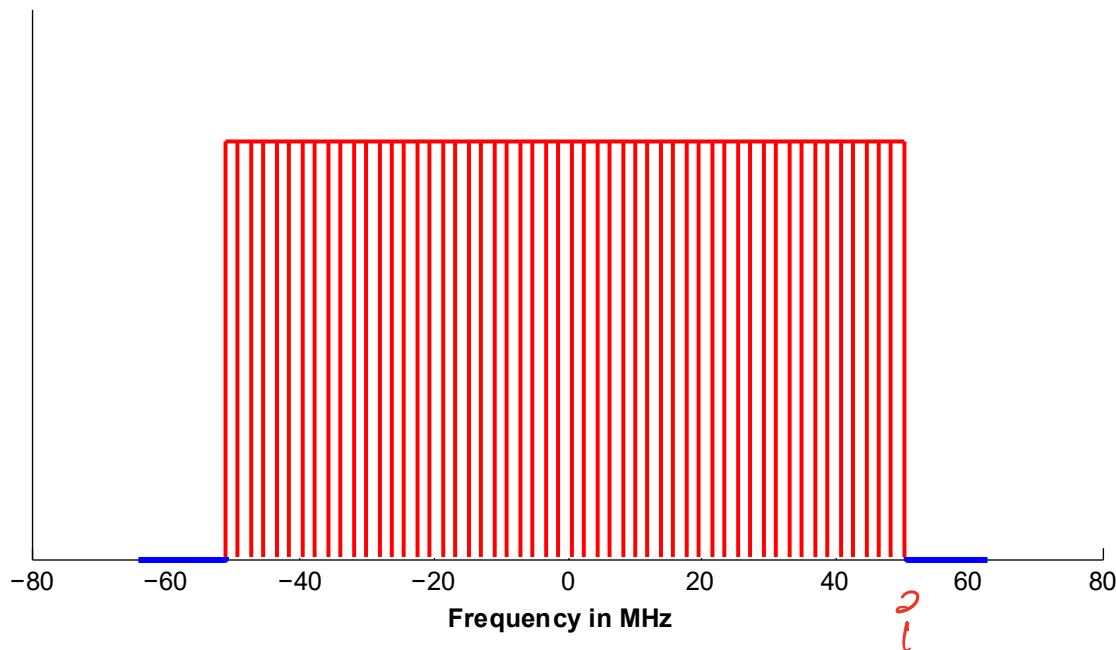
- ISI: Inter-Symbol-Interference limits performance



Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



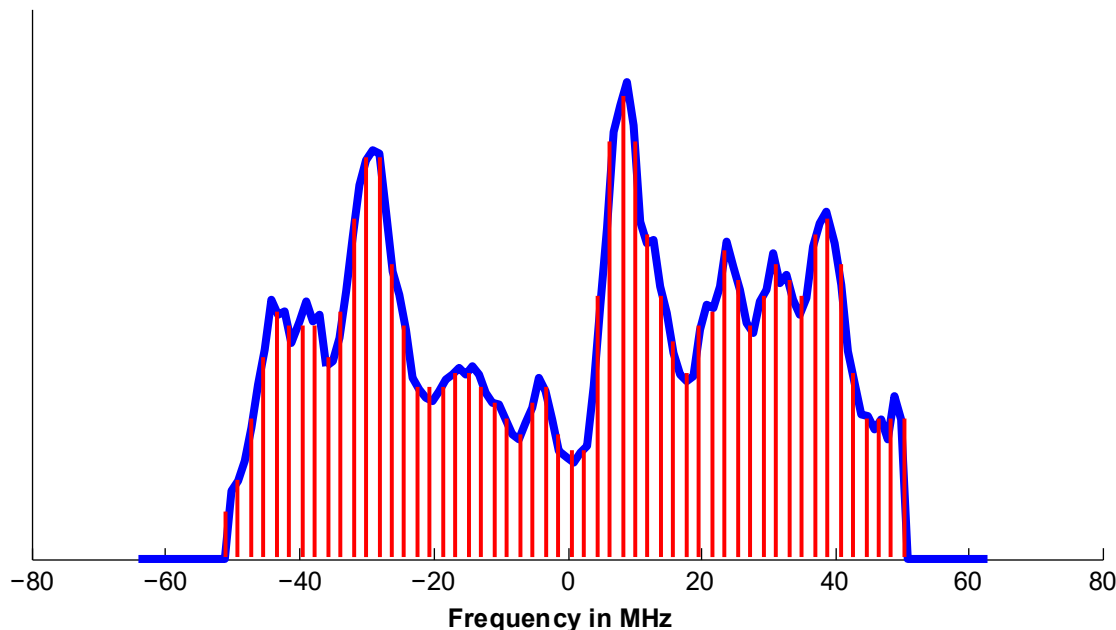
$$x(t) = \sum_i \underline{s_i[n]} e^{-j2\pi f_i t}$$

- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j 2 \pi f_i t}$$

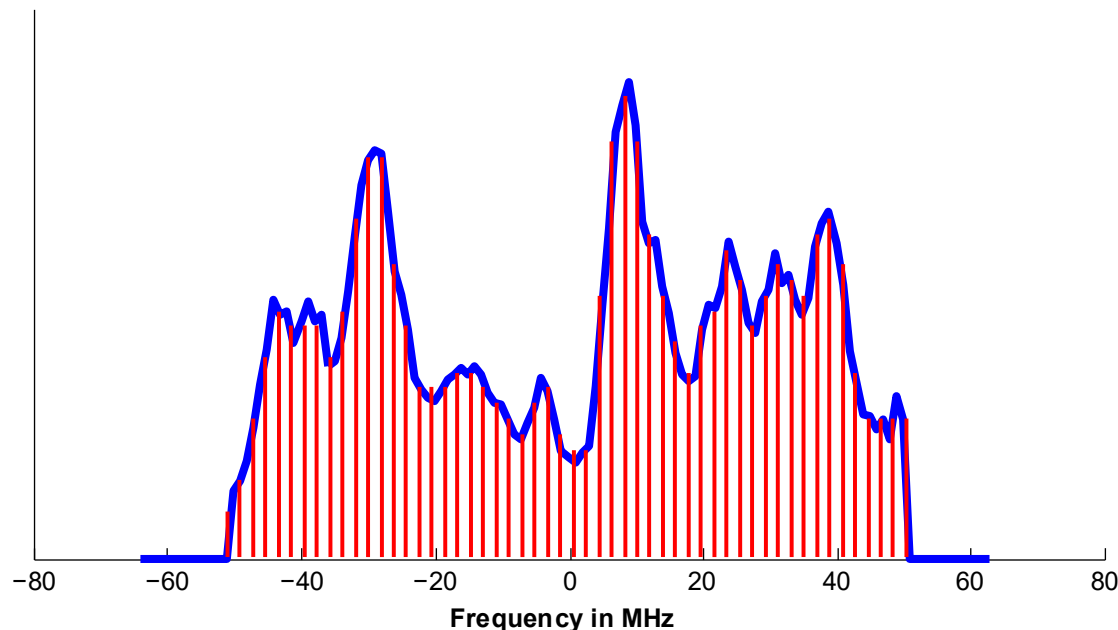
$$y(t) = \sum_i h_i s_i[n] e^{-j 2 \pi f_i t}$$

- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j 2 \pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j 2 \pi f_i t}$$

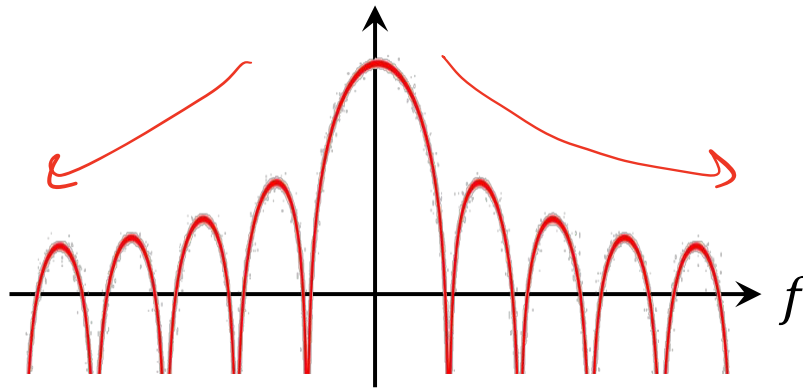
- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Not That Simple!

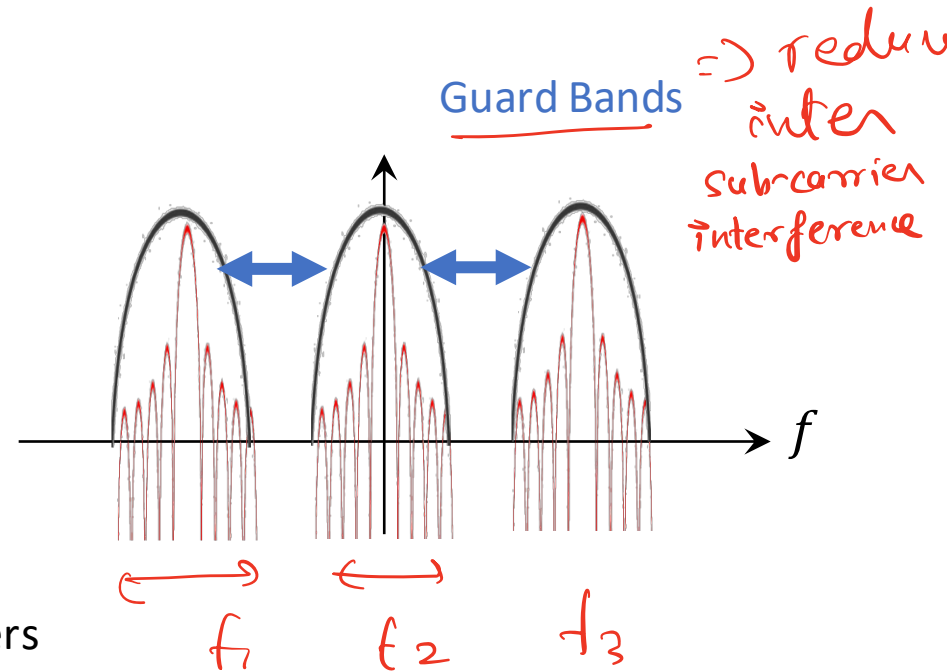
Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



- Significant Leakage between adjacent subcarriers
- Need Guard Bands → Very inefficient!

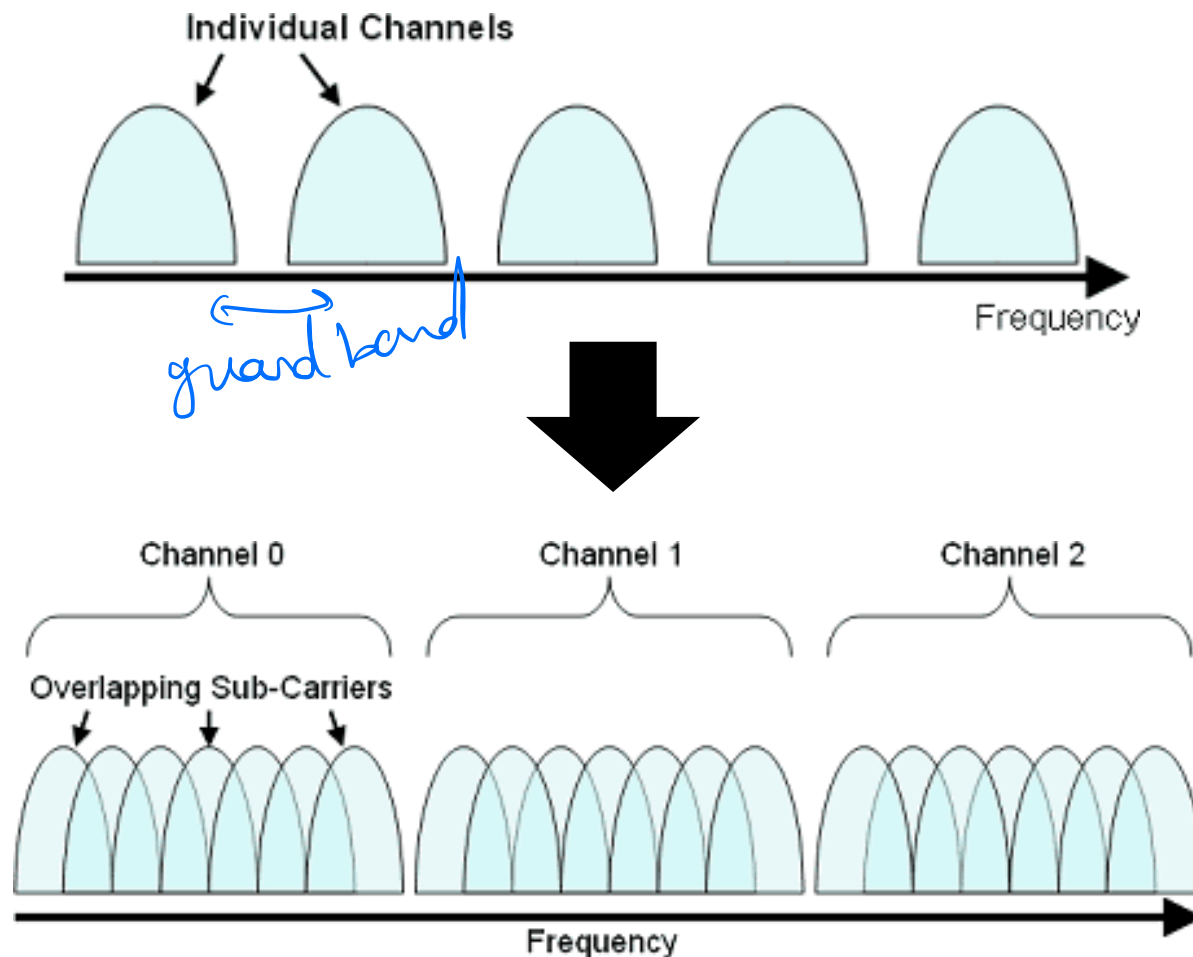


Solution: Make the Sub-Carriers Orthogonal

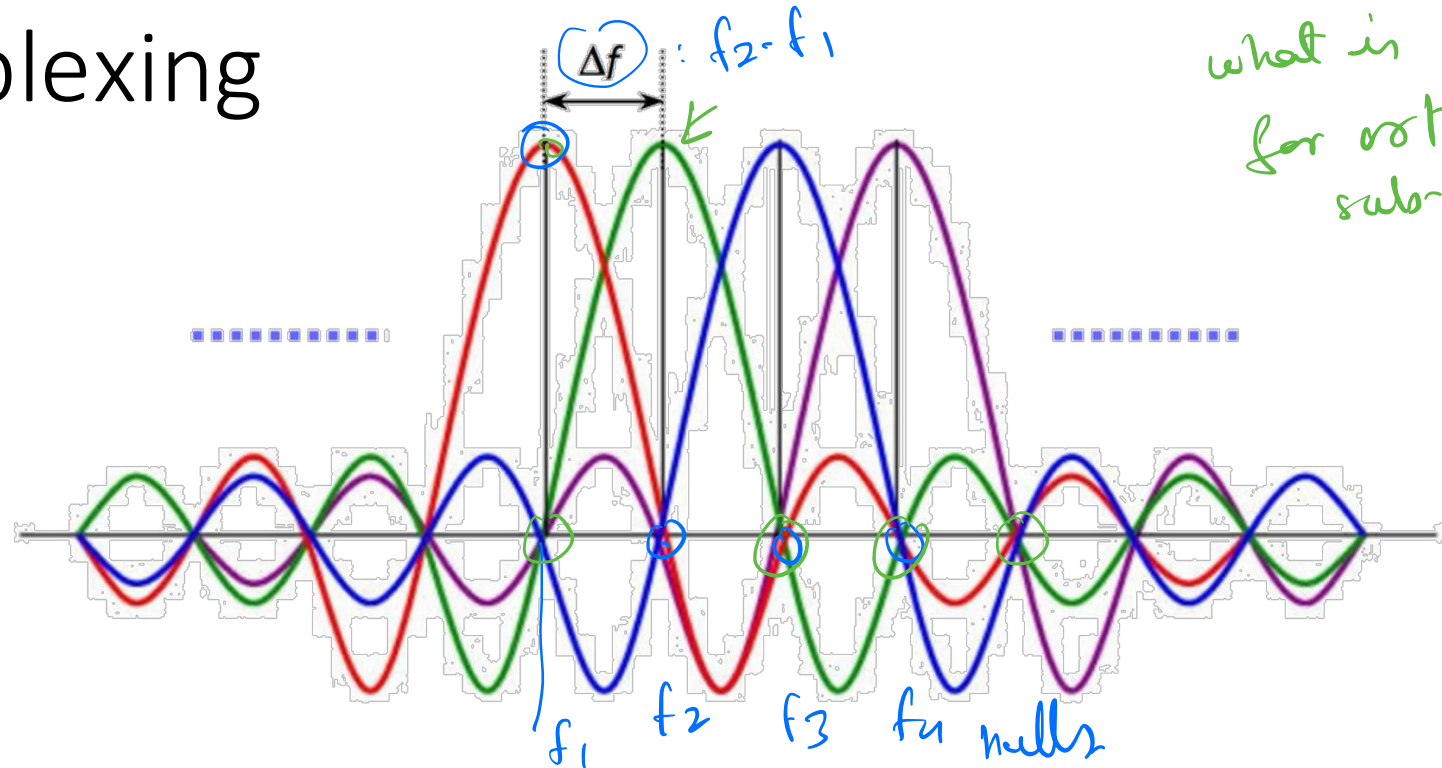
Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

Make the Sub-Carriers Orthogonal



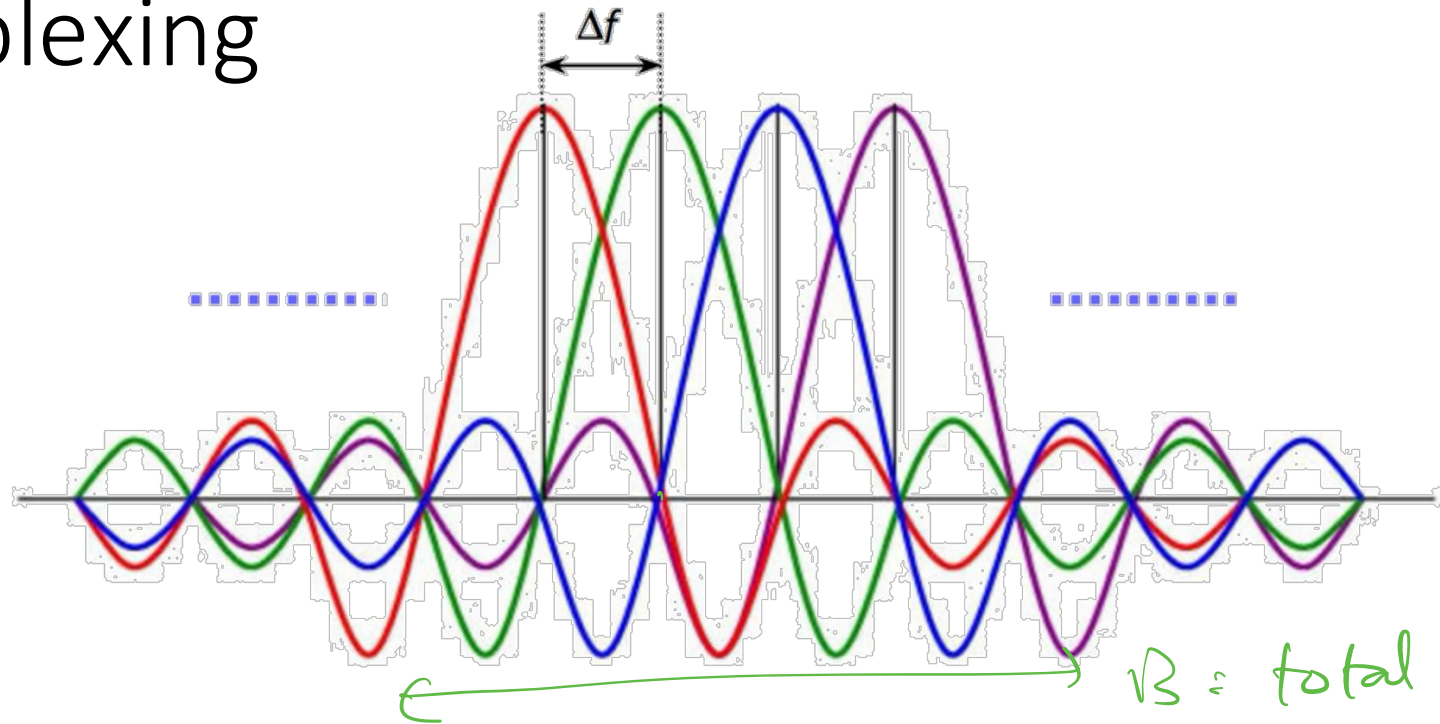
OFDM: Orthogonal Frequency Division Multiplexing



- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
- Subcarrier separation can be very small, for N subcarriers and bandwidth B :

$$\Delta f = \frac{B}{N}$$

OFDM: Orthogonal Frequency Division Multiplexing



- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
- Subcarrier separation can be very small, for N subcarriers and bandwidth B :

$$\Delta f = \frac{B}{N}$$

How to Achieve This?

OFDM: Orthogonal Frequency Division Multiplexing

Use DFT: Discrete Fourier Transform

$$\text{N-Point DFT: } X(f_i) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi f_i t}{N}}$$

Discrete $\rightarrow$ Discrete

$$x(t) \rightarrow \boxed{\text{DFT}} \rightarrow X(f)$$

$$\text{N-Point IDFT: } x(t) = \sum_{f_i=0}^{N-1} X(f_i) e^{j \frac{2\pi f_i t}{N}}$$

$$\begin{aligned} &\leftarrow x(f_0) e^{j \frac{2\pi f_0 t}{N}} = x(t) \\ &+ x(f_1) e^{j \frac{2\pi f_1 t}{N}} = x'(t) \\ &\vdots \end{aligned}$$

Send symbols in Frequency Domain

$X(f_i) = s[n] \rightarrow$ Compute and transmit $x(t)$ using IDFT

$\overbrace{s[n]}^{\text{BPSK}} \in \{+A, -A\}$

OFDM: Orthogonal Frequency Division Multiplexing

Send symbols in Frequency Domain

$X(f_Q)$ = $s[n]$ $\rightarrow$ Compute and transmit $x(t)$ using IDFT

- $N_{\text{subcarrier}}$ $\rightarrow$ IDFT of length N $\Rightarrow$ *N time domain samples*
- Symbols $s[n]$ can come from any modulation: BPSK, QPSK, QAM...
- $x(t)$ is complex $\rightarrow$ need I & Q $\rightarrow$ No point using PAM or ASK ...
- OFDM Symbol: N samples of $x(t)$ generated from the same modulated symbols using IDFT.
- OFDM Symbol Time: $T = N/B$ where B is the bandwidth. *$\Rightarrow$ total bandwidth of OFDM symbol*
- OFDM Frequency Bin Width: $\Delta f = 1/T = B/N$



$$T_{\text{ofdm}} = \frac{1}{\Delta f}$$

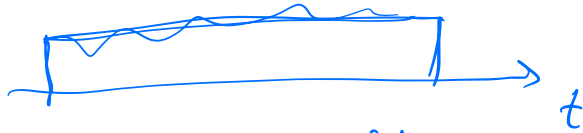


$$1/\Delta f$$



$$x'(t)$$

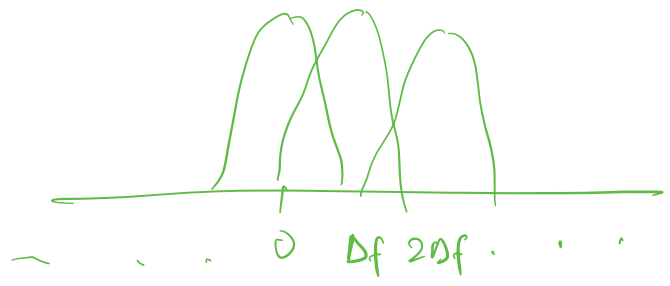
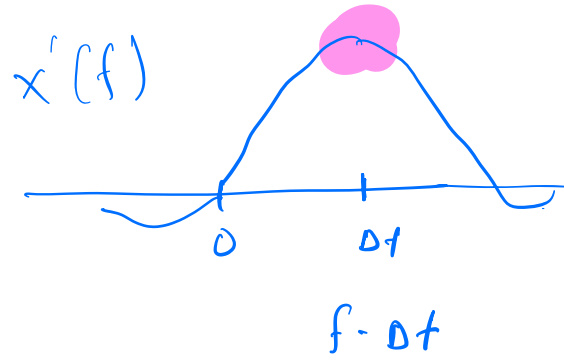
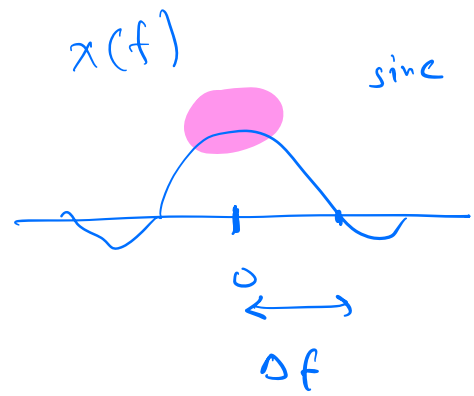
complex



$$x(t) \times e^{j2\pi \Delta f t}$$

$$x(t) + x'(t) \dots$$

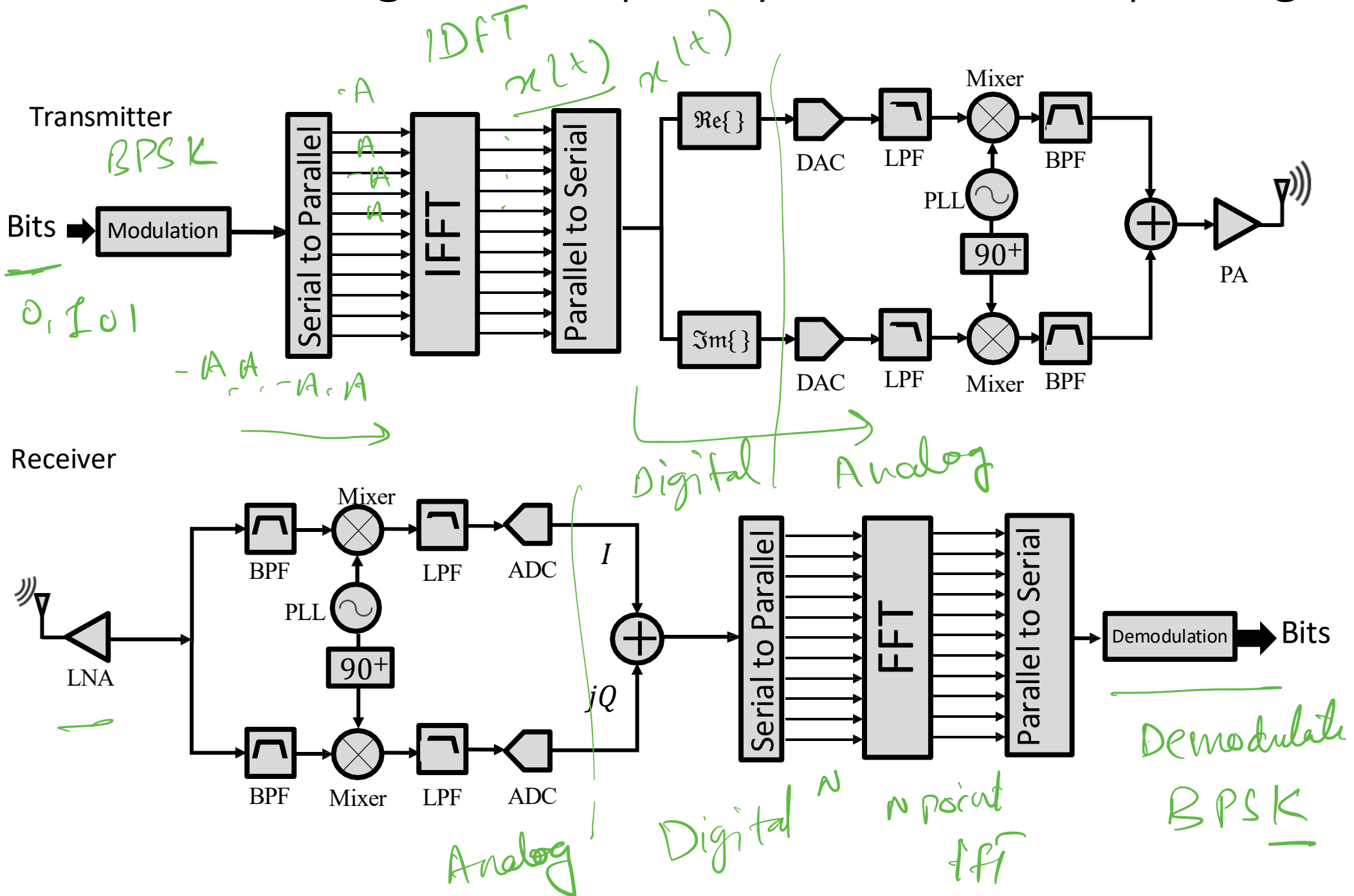
$$T_{\text{ofdm}} = \frac{1}{\Delta f} = \frac{N}{B}$$



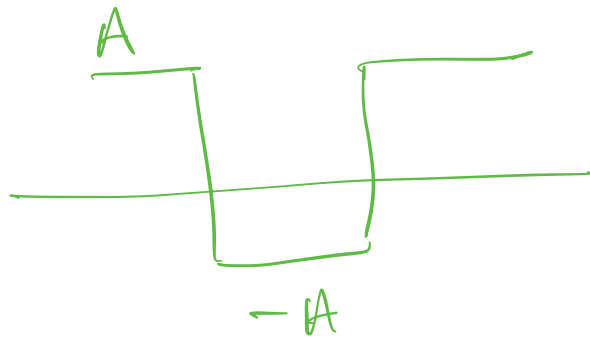
N over bandwidth B

$$\Delta f = \frac{B}{N}$$

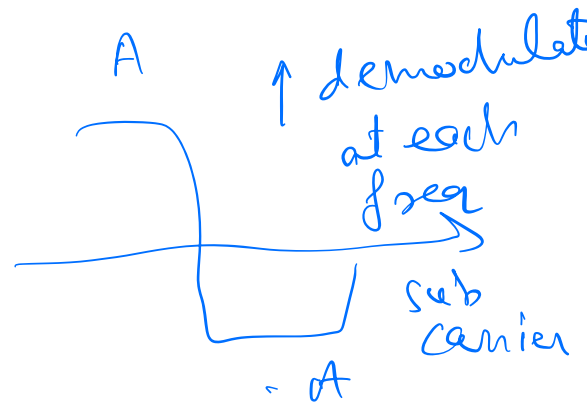
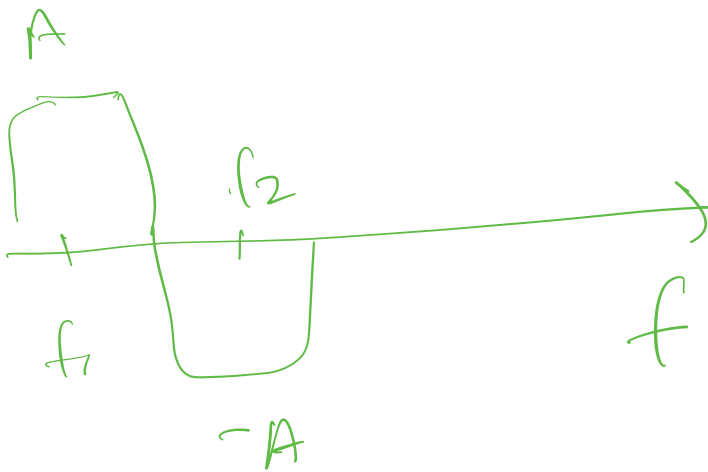
OFDM: Orthogonal Frequency Division Multiplexing



time domain processing



OFDM: freq domain processing

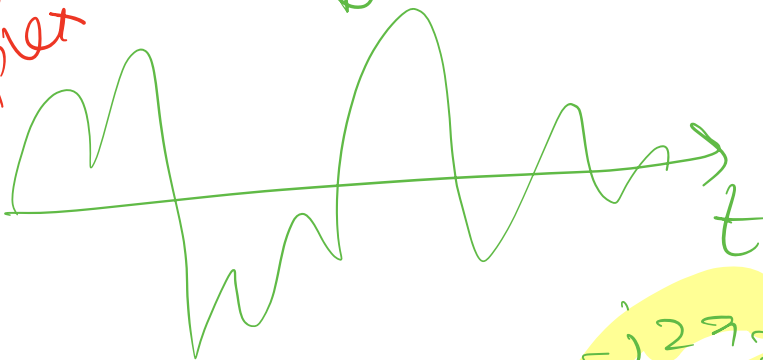


↑ FFT at receiver

→ 1 Q mixer

random complex

↓ IFFT

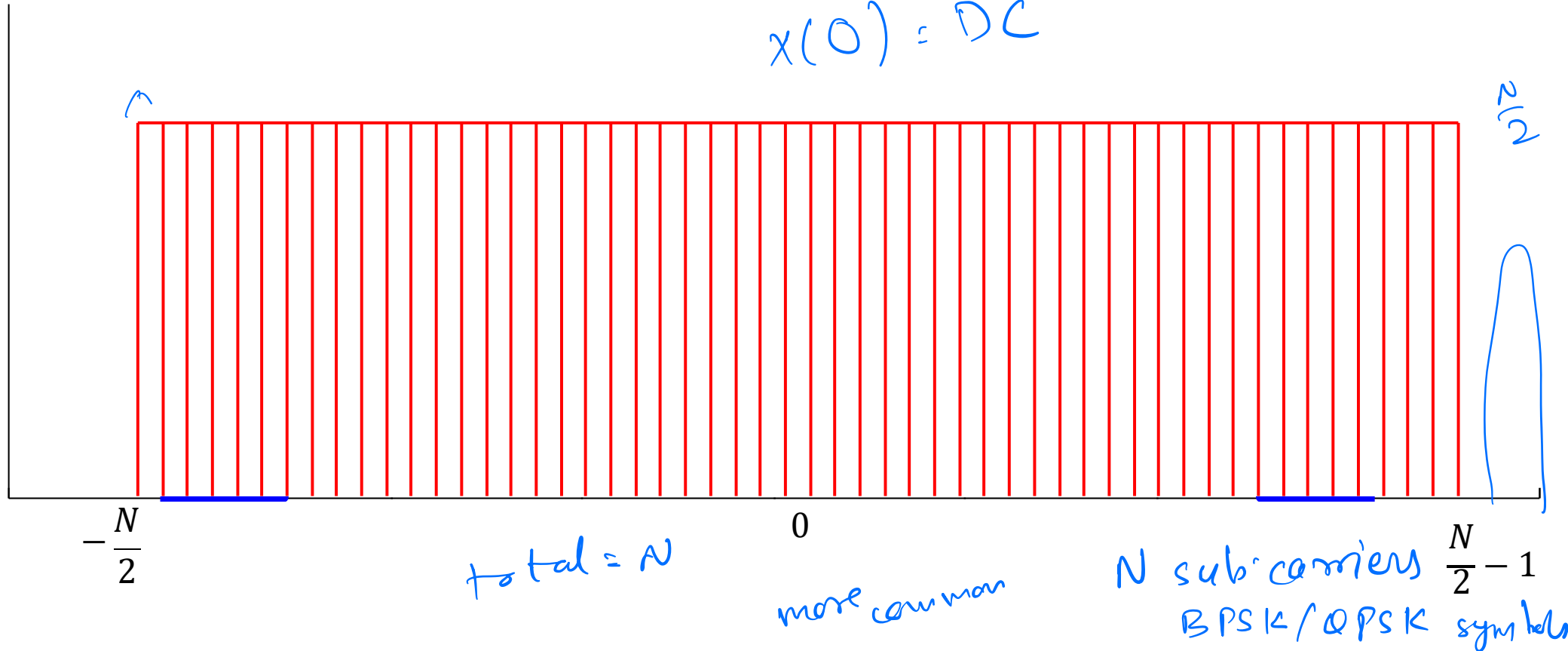


$$\frac{1}{N} \sum_i X(f_i) e^{-j2\pi f_i t}$$

↑ add multiple symbols

OFDM Symbol in Frequency Domain

$$x(0) = DC$$



- FFT can be represented 0 to N - 1 or -N/2 to N/2 - 1.
- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC

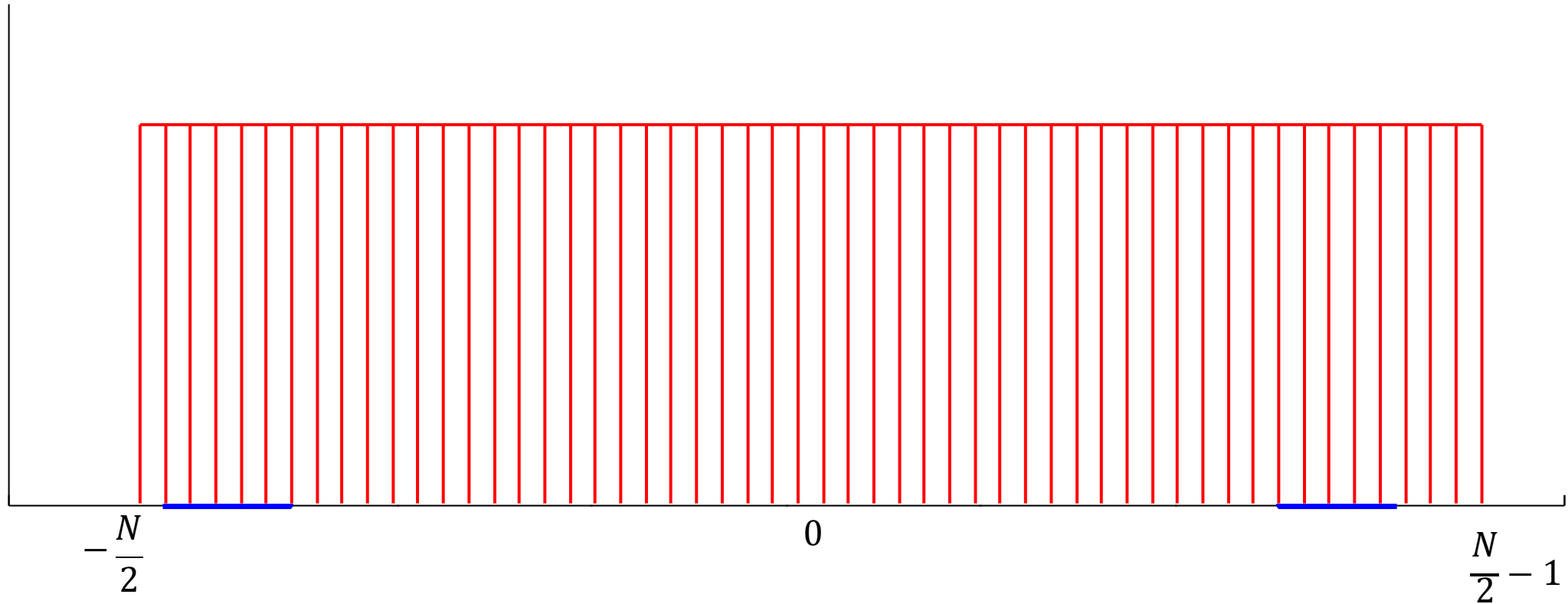
$\downarrow$ IFFT

$\Rightarrow$ 1 OFDM symbol

$$\underline{X(0)} = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi 0 t}{N}} = \frac{1}{N} \sum_{t=0}^{N-1} x(t) = \underline{DC}$$

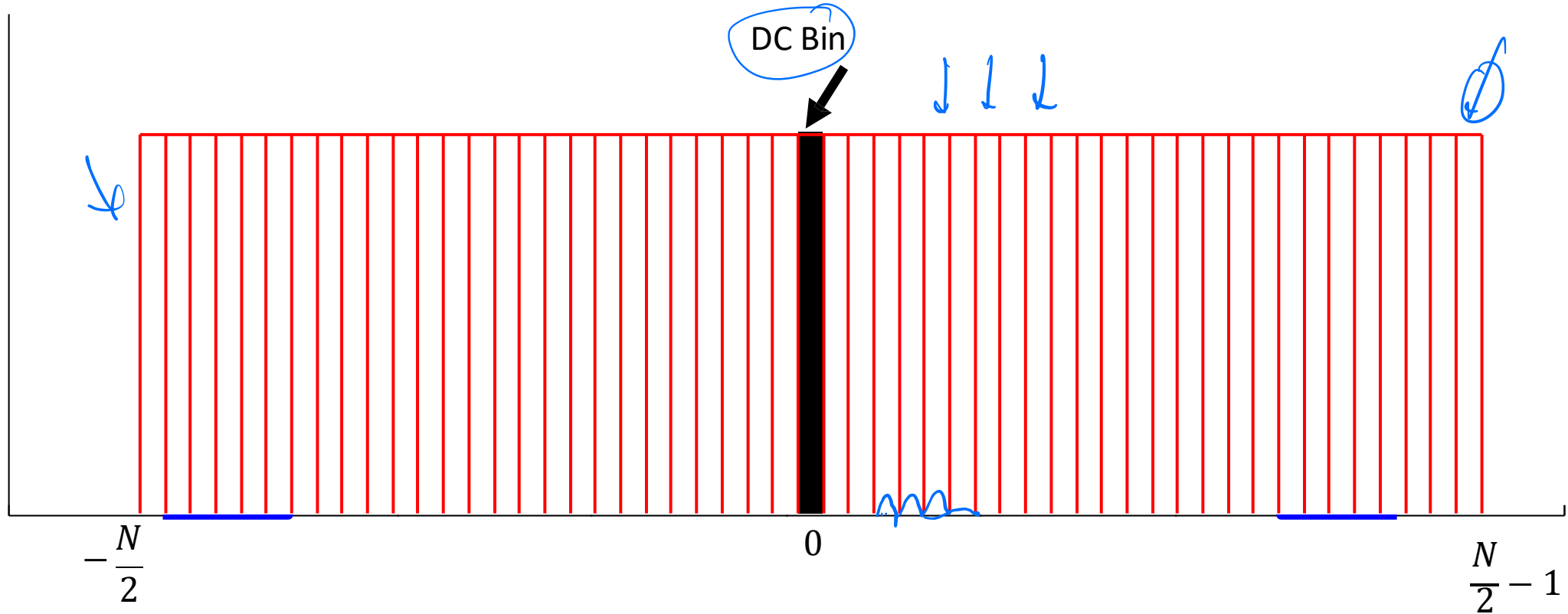
DFT

OFDM Symbol in Frequency Domain



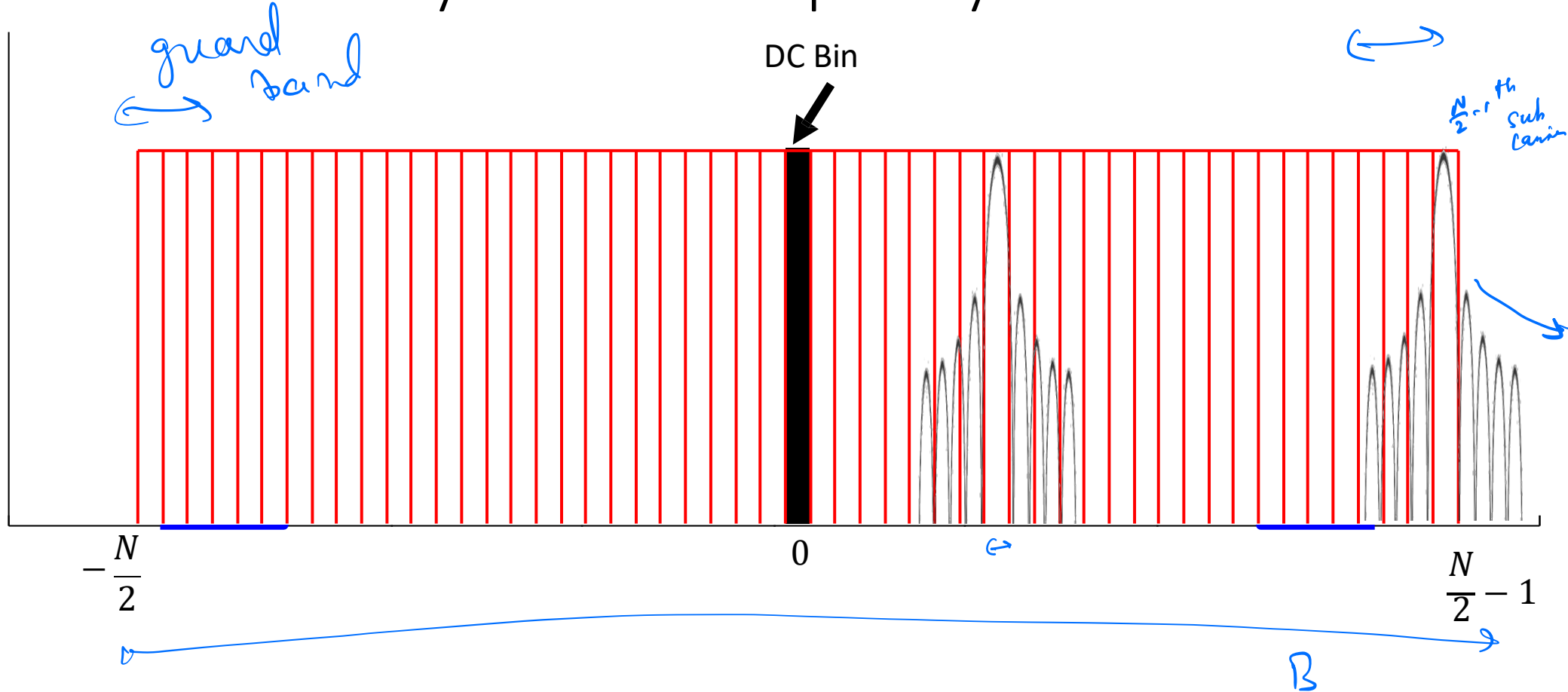
- FFT can be represented 0 to $N - 1$ or $-N/2$ to $N/2 - 1$.
- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC
- DC of the circuits corrupts bits sent on the 0 bin $\rightarrow$ Do not use 0 bin

OFDM Symbol in Frequency Domain



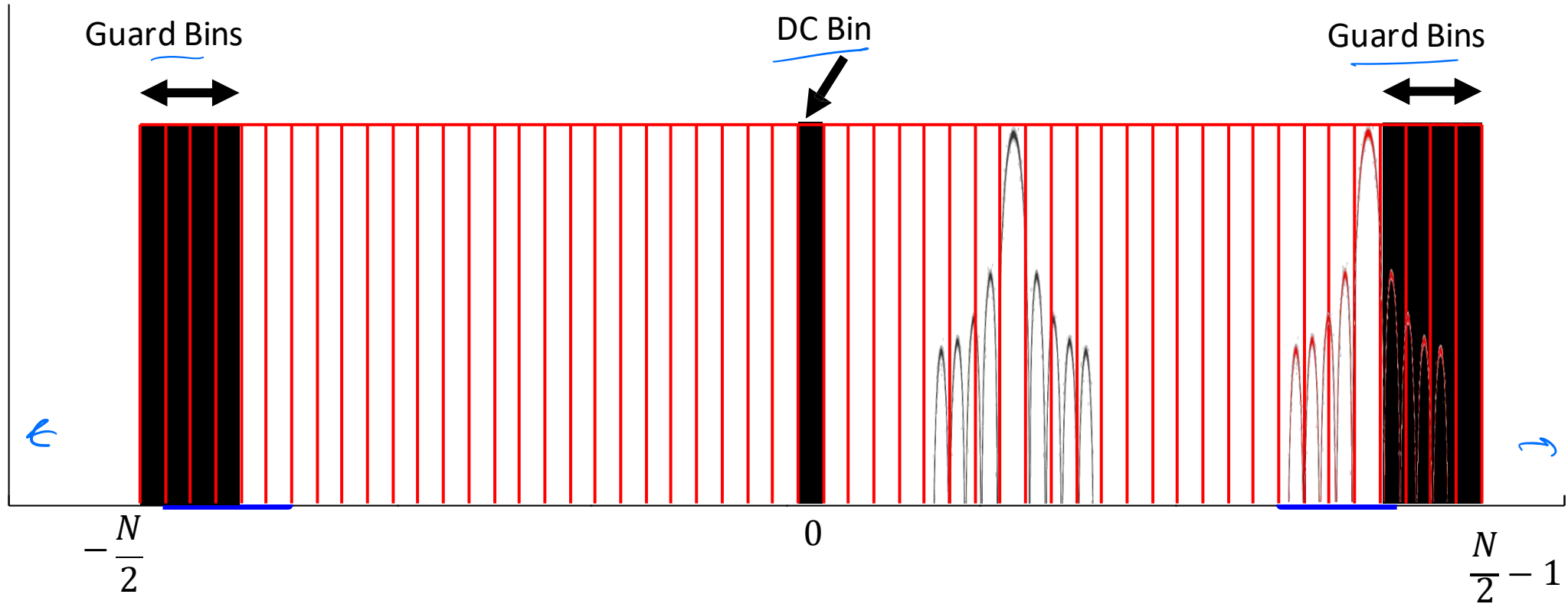
- FFT can be represented 0 to $N - 1$ or $N/2$ to $N/2 - 1$.
- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC
- DC of the circuits corrupts bits sent on the 0 bin $\rightarrow$ Do not use 0 bin

OFDM Symbol in Frequency Domain



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there

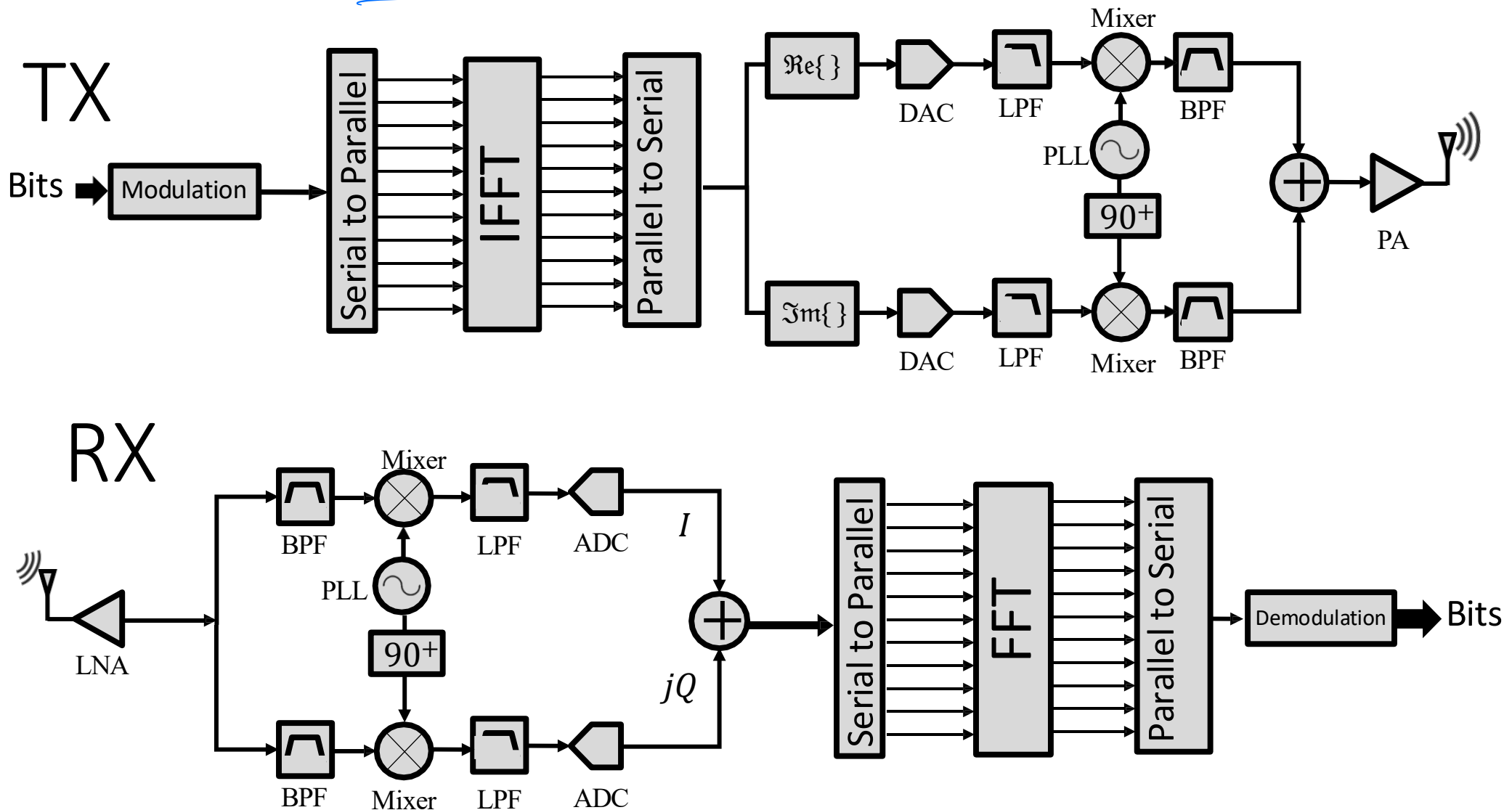
OFDM Symbol in Frequency Domain



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there
- Reduce Number of Guard band from N to 2 → Very Spectrally Efficient

OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers

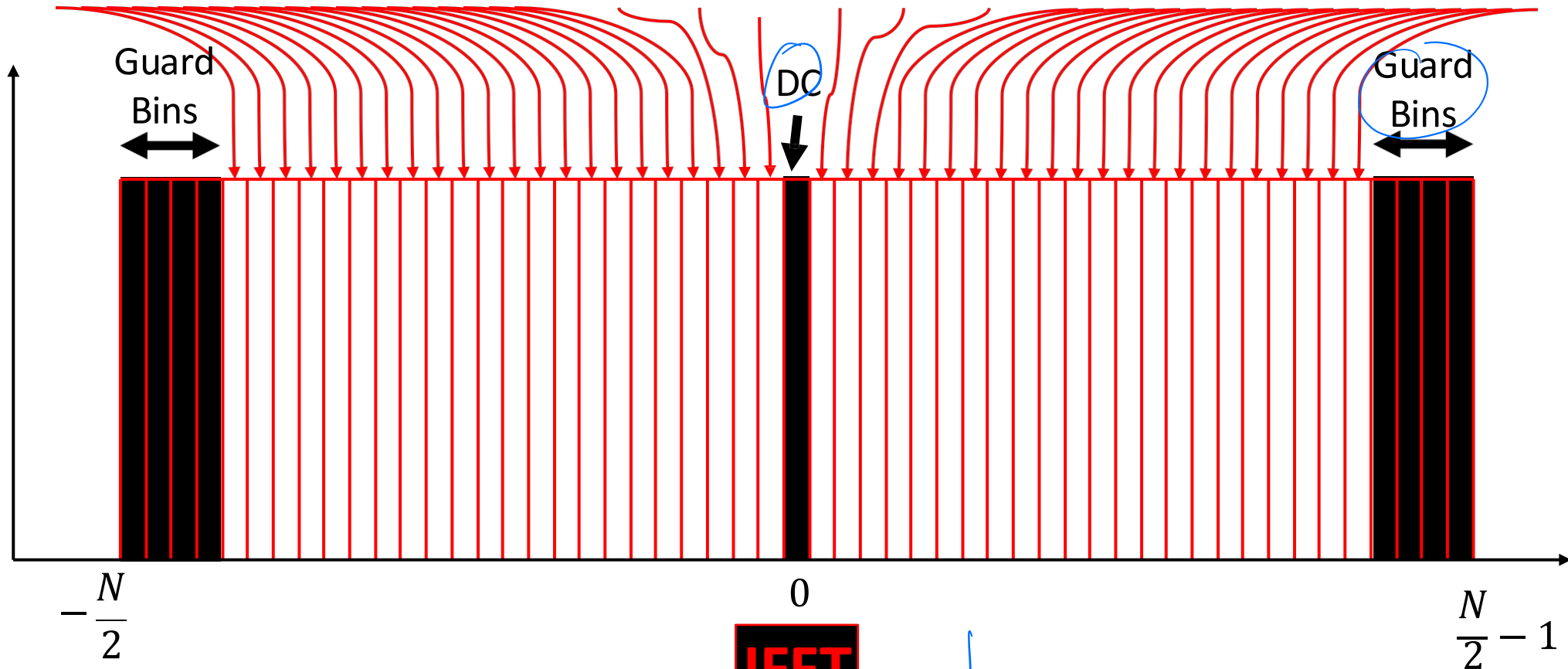


OFDM Symbol

Bits: 1 0 1 0 1 0 0 0 1 1 0 1 1 0 0



..., +1, -1, +1, -1, +1, -1, -1, -1, +1, +1, -1, +1, +1, -1, -1, ...

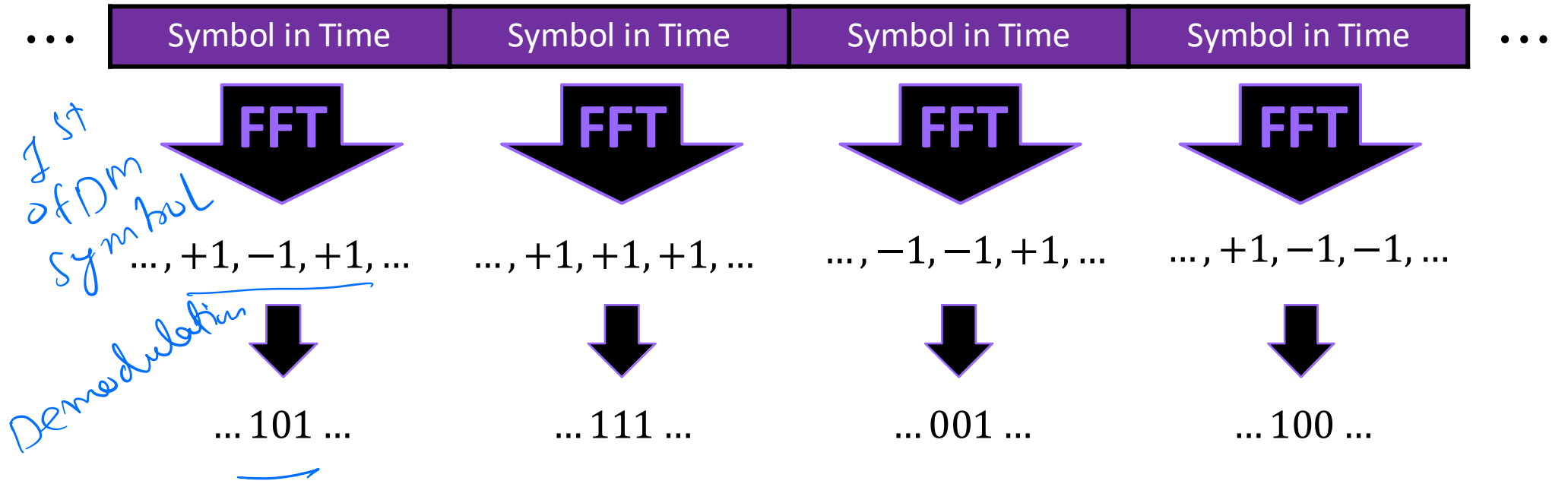


Symbol in Time



Receiver
Processing

OFDM Symbol "ideal"



Not That Simple

→ noise

→ Non-linearity

→ noise

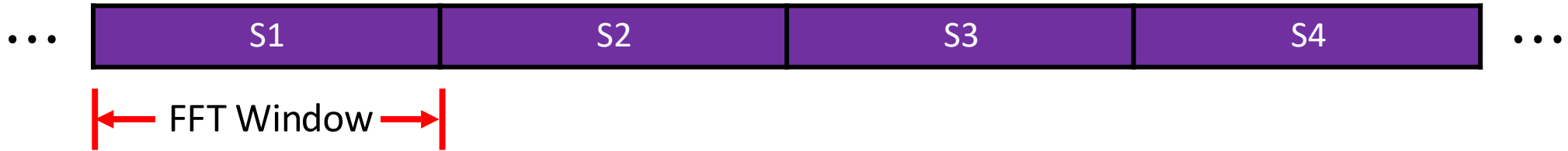
→ Channel equaliz

→ freq offset

→ time offset

Time window

OFDM Symbol



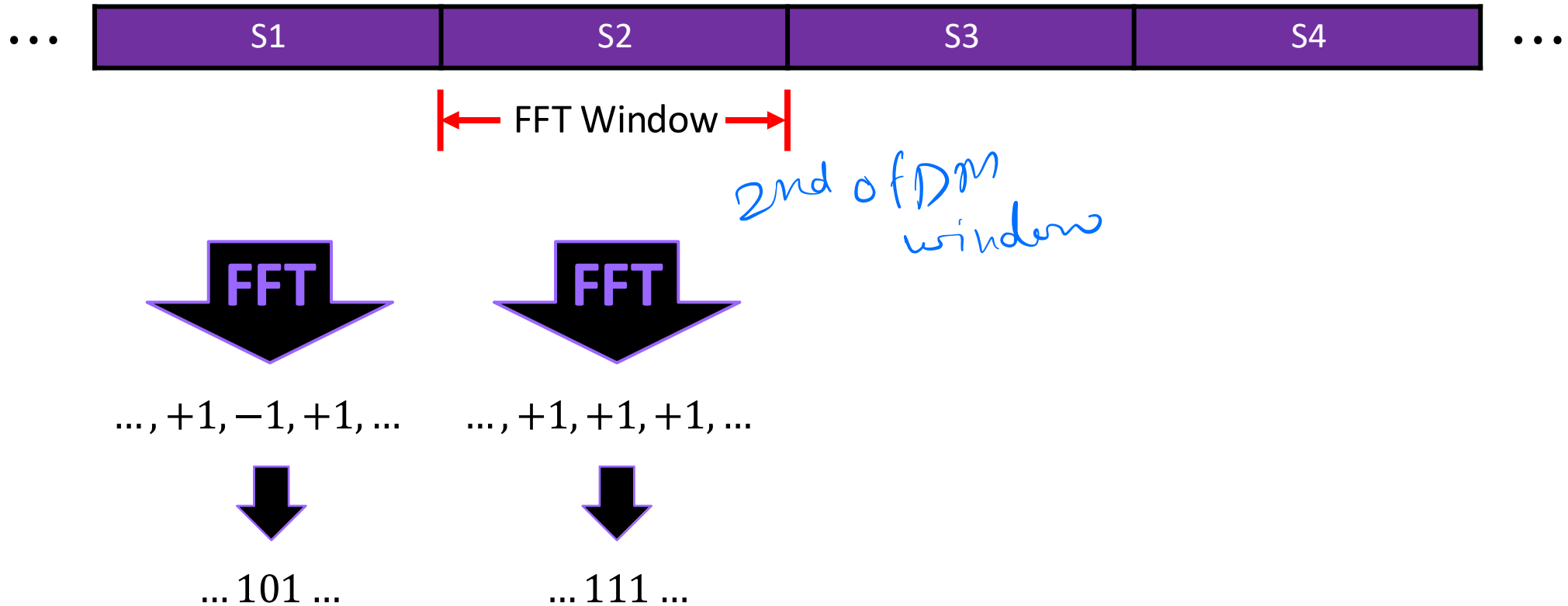
..., +1, -1, +1, ...



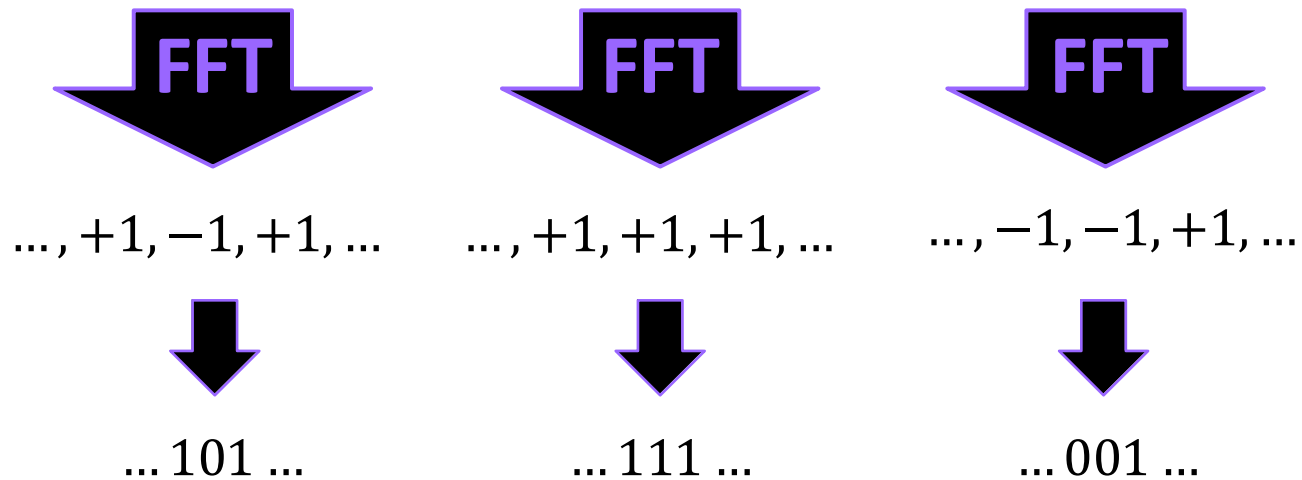
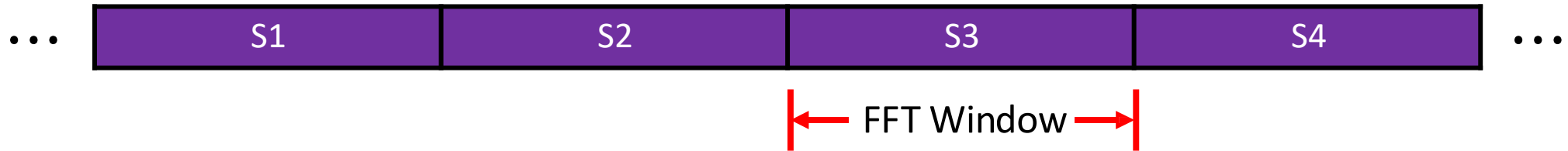
... 101 ...

↓
perfect

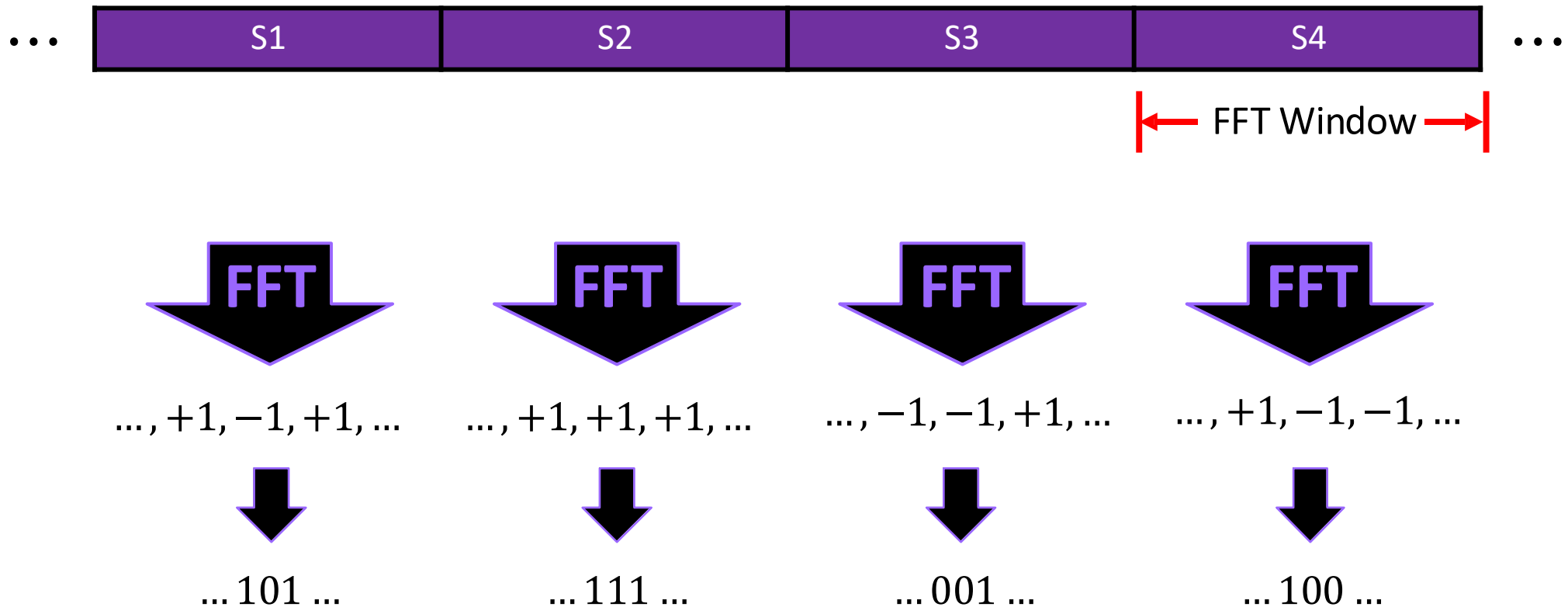
OFDM Symbol



OFDM Symbol

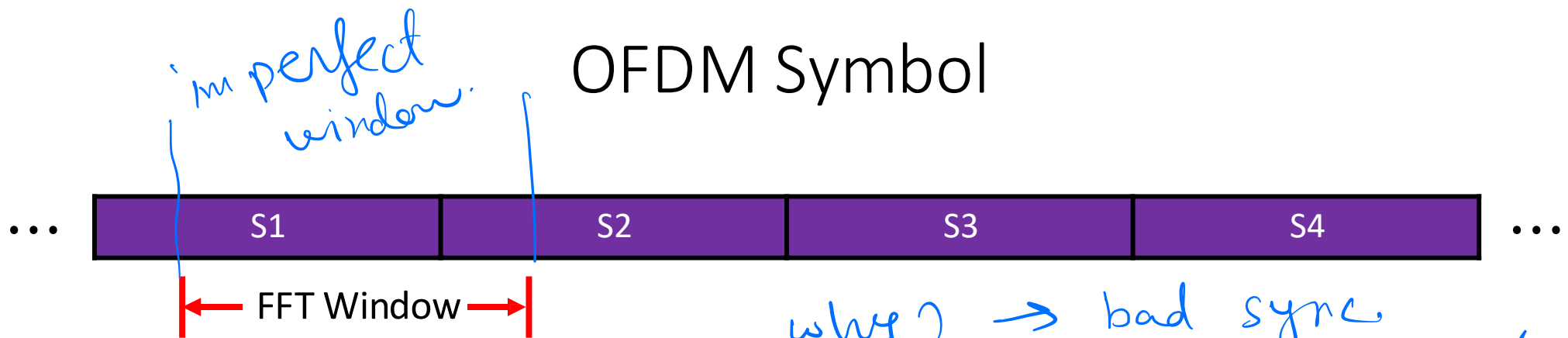


OFDM Symbol



Assumes FFT window is perfectly aligned with symbol boundaries

OFDM Symbol



..., $+0.5 + 1i$, $-0.7 + 0.3i$, ...

✗ Cannot decode!

why? → bad sync
→ need to estimate
symbol
boundary

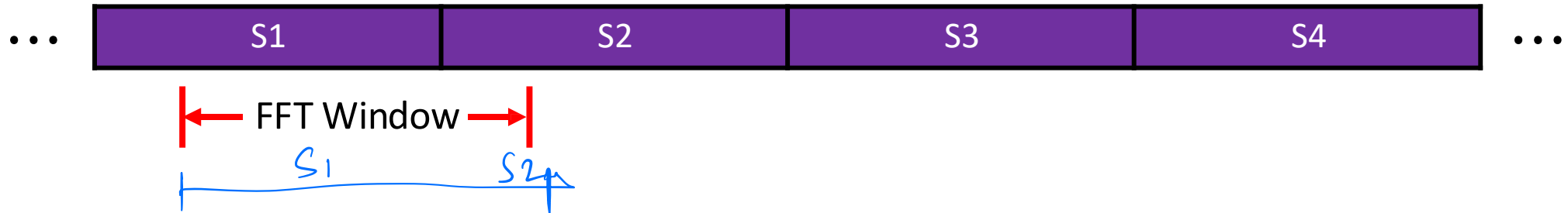
→ Multipath channel

→ ADC / DAC
time offset

FFT window is misaligned with symbol

Subcarriers are no longer orthogonal.

OFDM Cyclic Prefix

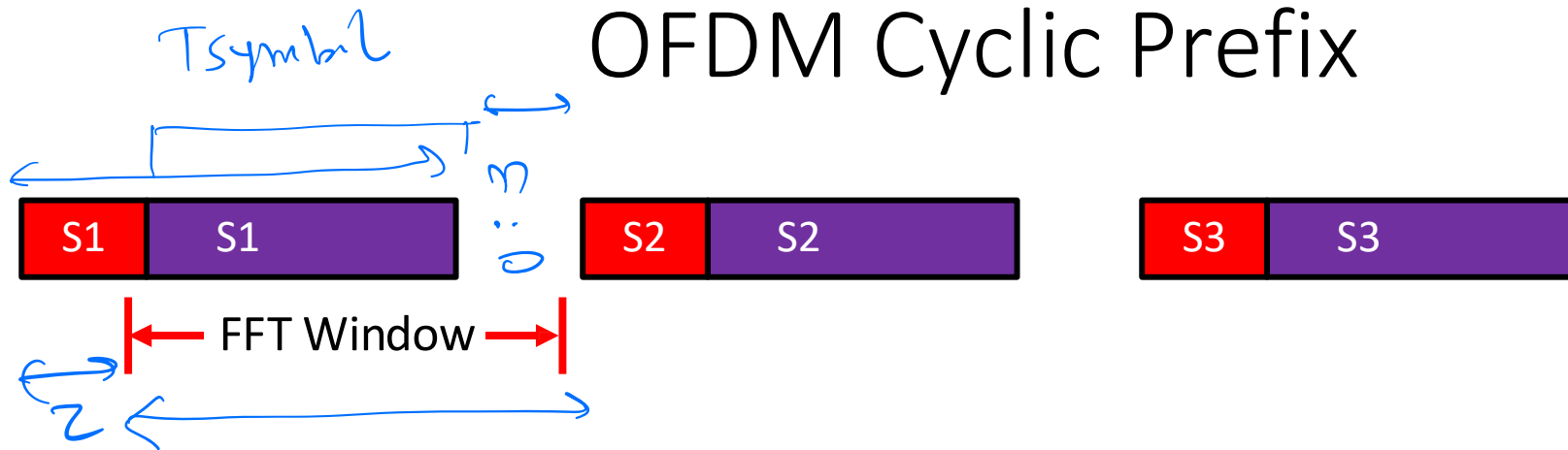


- DFT (FFT) assumes time samples are periodic of period N
- Circular Shift before taking FFT:

$$x[t] \rightarrow X[f]$$

$$\underline{x[t - \tau \bmod N]} \rightarrow \underline{X[f] e^{-j \frac{2\pi f \tau}{N}}}$$

OFDM Cyclic Prefix



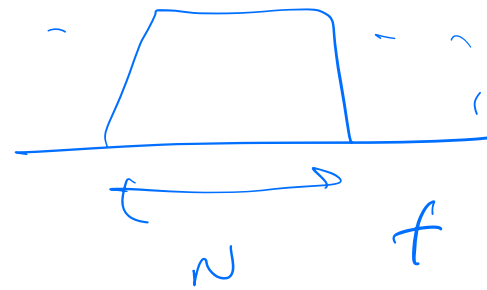
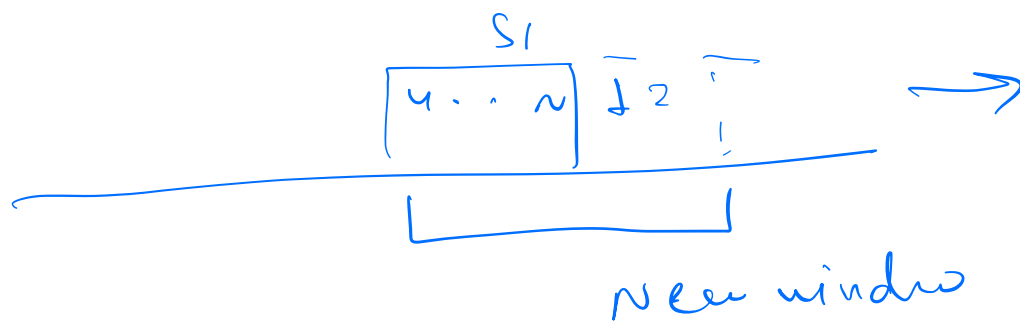
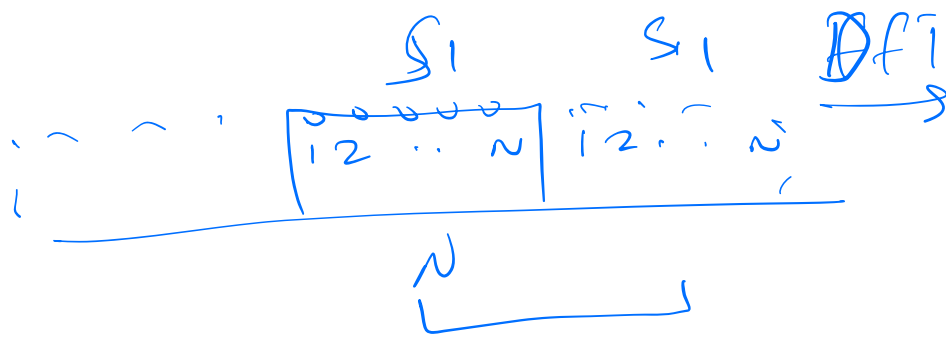
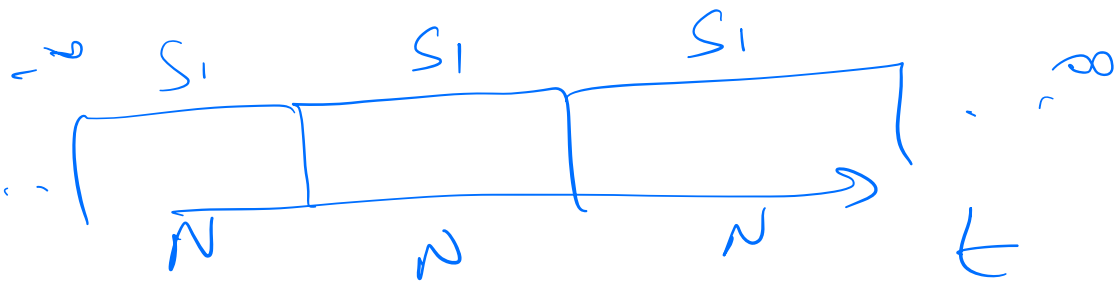
- DFT (FFT) assumes time samples are periodic of period N

- Circular Shift before taking FFT:

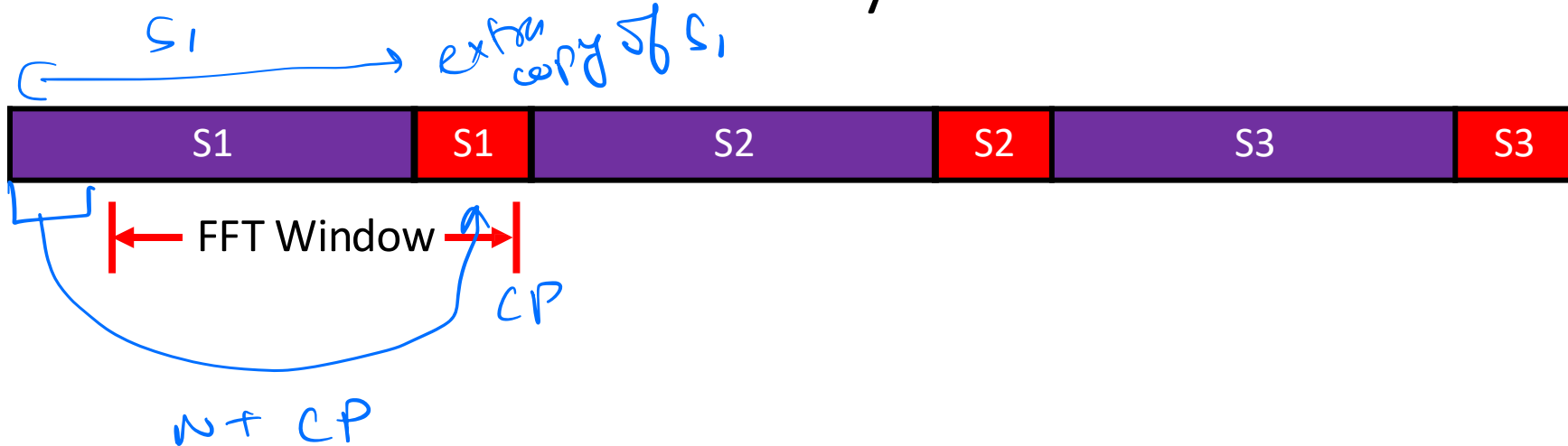
$$\underline{x[t]} \rightarrow X[f]$$

$$x[t - \tau \bmod N] \rightarrow X[f] e^{-j \frac{2\pi f \tau}{N}}$$

for
factor to be compensated by channel equalizer
extension phase factor



OFDM Cyclic Prefix



$$0 - N - 1$$

$$1 - N + 1$$

$$CP - N + CP - 1$$

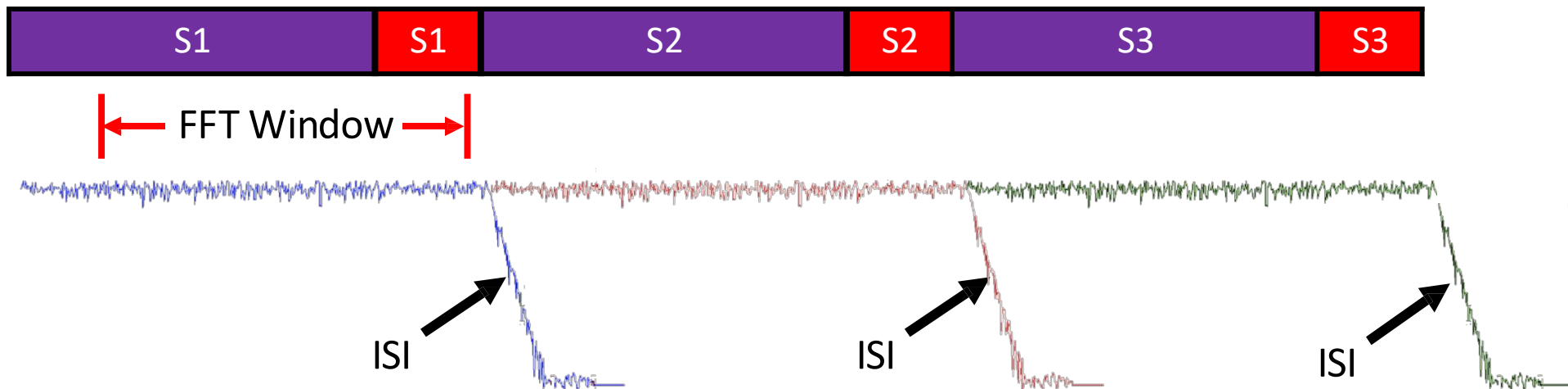
- Even if FFT window is misaligned, CP ensures that all samples come from the same symbol → Orthogonality is preserved!

- Simple Phase Shift → Can be corrected by lumping with channel $H[f]$
- Cyclic Prefix can be created by:
 - Take first few samples and append them to end of symbol.
 - Take last few samples and prefix them to beginning of symbol.

OFDM Cyclic Prefix

Cyclic Prefix:

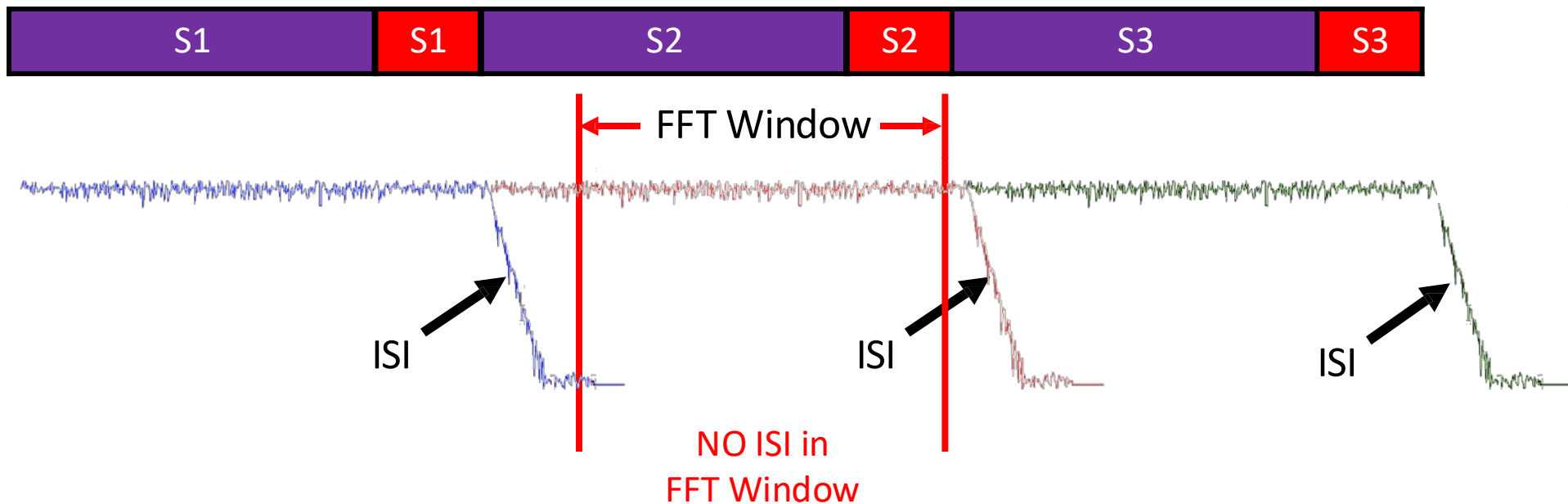
- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

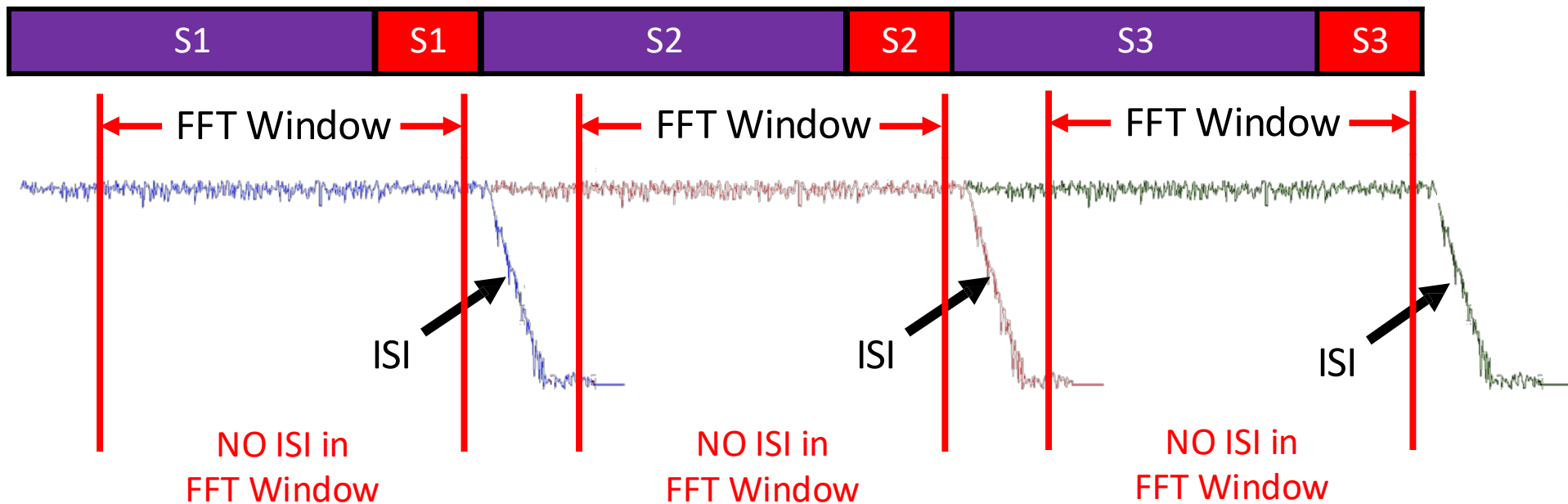
- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

 Preserves orthogonality by allowing some misalignment in FFT Window

 Deals with Inter-Symbol-Interference

 Overhead: Send $CP + N$ samples for every N samples

$$\text{Overhead} = \frac{CP}{CP + N}$$

e. g. WiFi 802.11n: $N = 64$, $CP = 16 \rightarrow \text{Overhead} = 20\%$

e. g. LTE: $N = 1024$, $CP = 72 \rightarrow \text{Overhead} = 6.5\%$

OFDM Cyclic Prefix



- Cyclic prefix is not a bullet proof solution.
- Can still end up misaligned!
- Need a way to ensure we detect the beginning of the packet correctly.
- If we do, CP will ensure that even if we are not accurate, we can still decode.

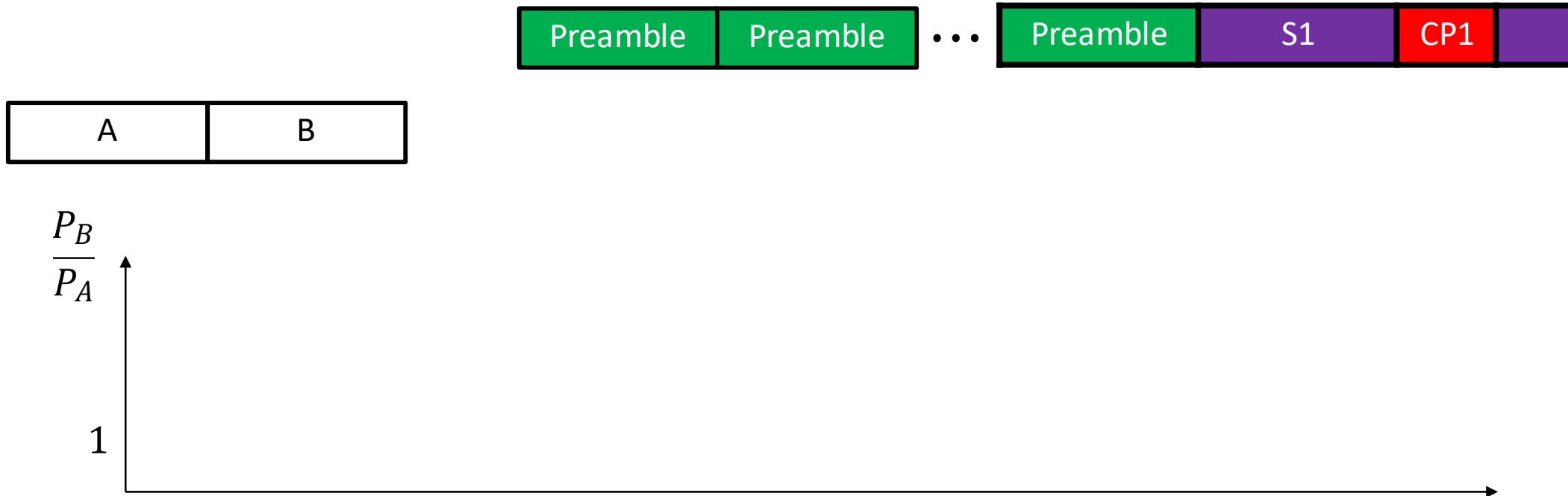
OFDM Packet Detection

- Detect Beginning of packet to make sure we are within the CP
- Send Training Sequence: Preamble Symbols
- Preamble Symbols: Known Symbol Repeated at the beginning of packet



- Generally, 2 copies of preamble (LTS – Long Training Sequence) used in WiFi + 0.5 copy as cyclic prefix

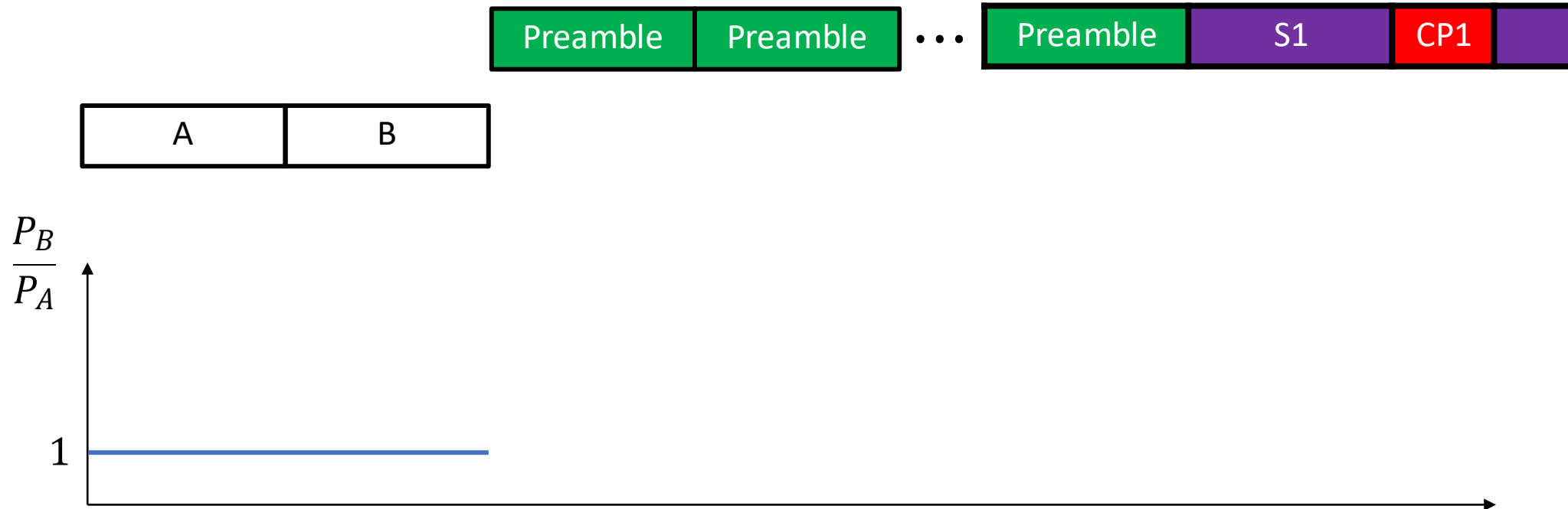
OFDM Packet Detection: Sliding Window



- Two windows of L ($2N$) samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

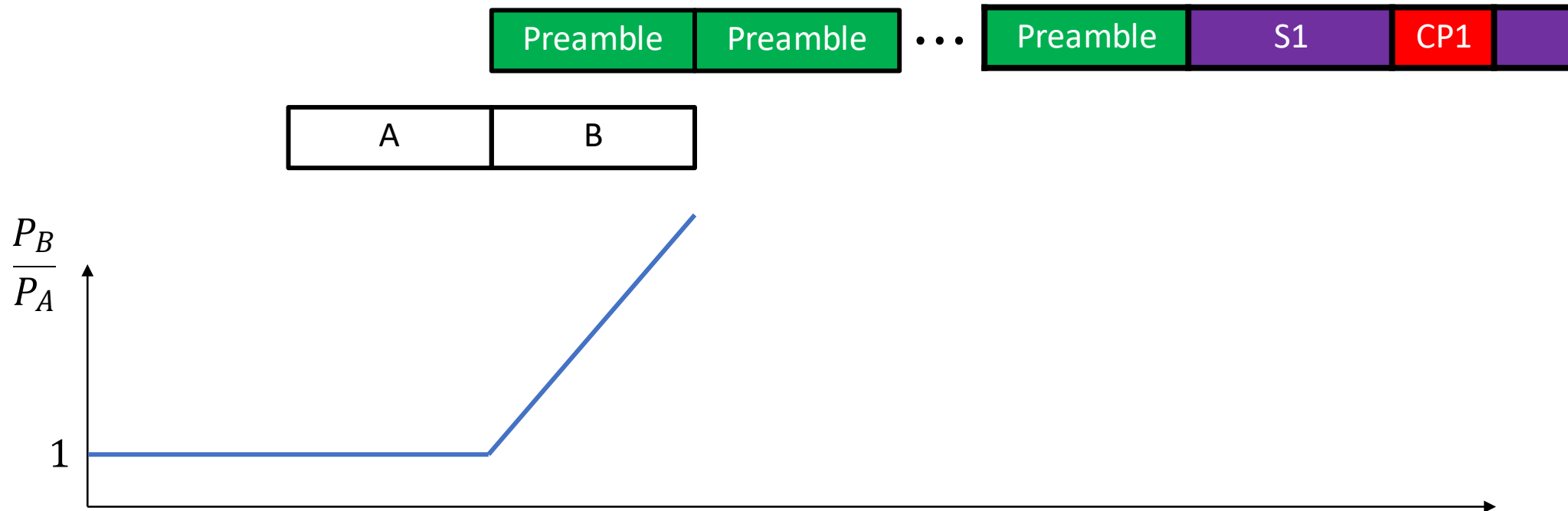
OFDM Packet Detection: Sliding Window



- Two windows of L ($2N$) samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

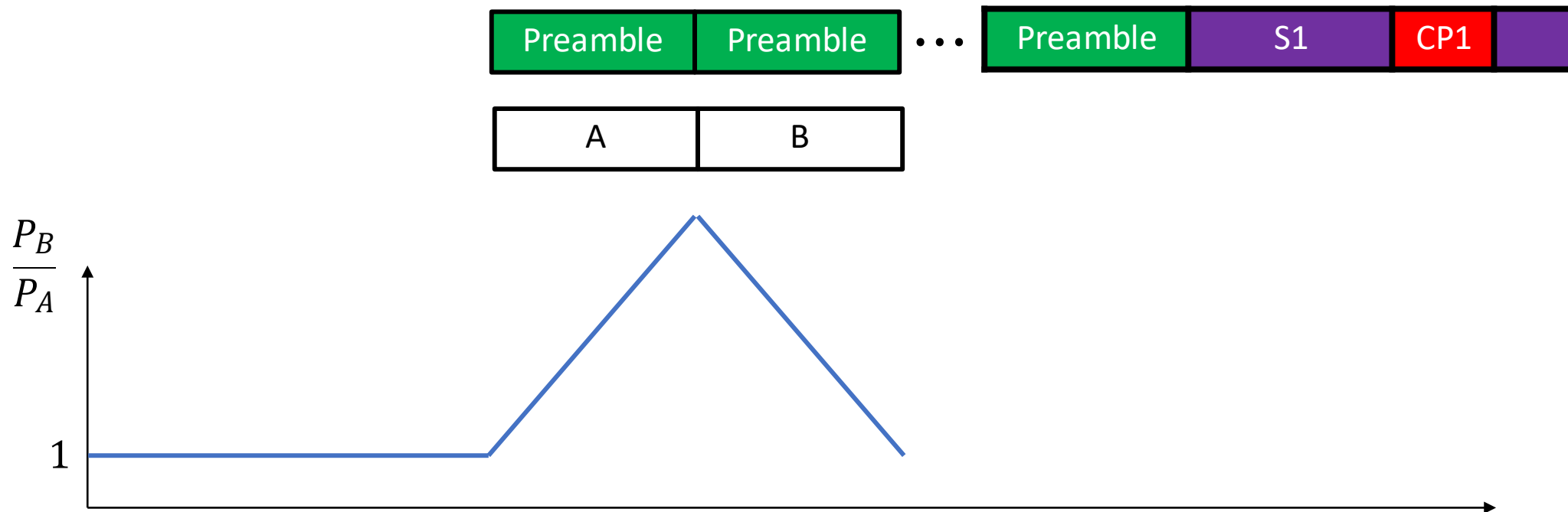
OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

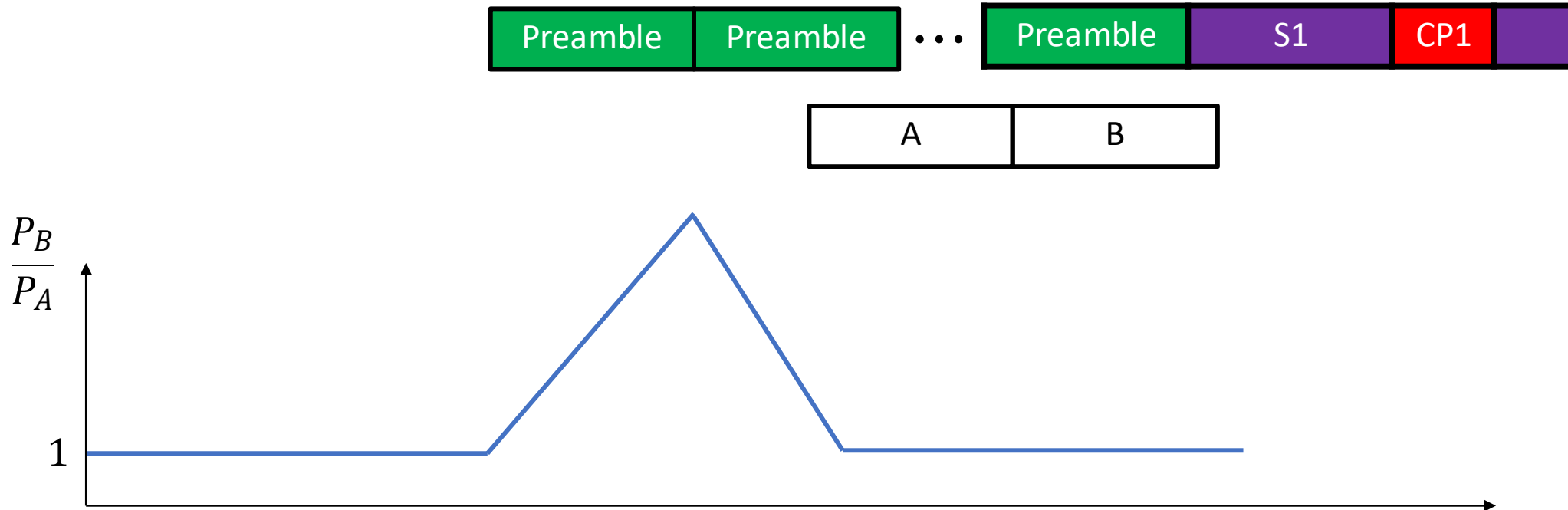
OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

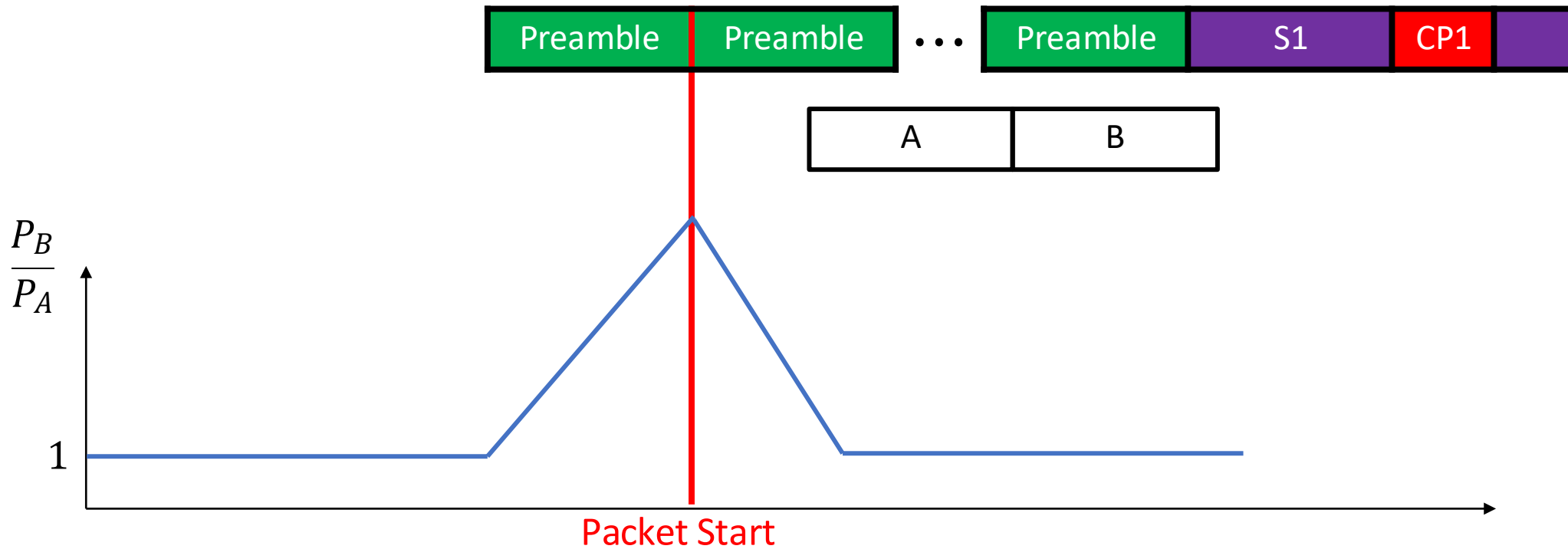
OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$